

From Agriculture to Services: Development without Industrialization

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*Preliminary - please do not circulate*¹

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Abstract

This paper studies the long-run consequences of developing without industrialization. Countries with higher historical peaks in manufacturing employment grow faster, accumulate more tertiary education, experience faster fertility declines, and develop more skill intensive service sectors. The same pattern appears within the United States: counties with larger manufacturing booms later developed service occupations with higher educational content. I interpret these facts through a unified growth model with structural change, endogenous fertility, education, and heterogeneous producer services. In the model, manufacturing creates demand for high-skill producer-service tasks that build the skill base and organizational frontier of the service economy that follows. Calibrated to the United States over 1850–2019, the model explains 31 percent of cross-country log GDP per capita gaps, 60 percent of GDP per capita growth differences, 53 percent of tertiary education differences, and 39 percent of cumulative population growth differences. Countries can move into services without industrializing, but the service economy they build is less skill intensive and grows more slowly.

Keywords: structural change, service led growth, industrialization, unified growth, fertility, human capital.

JEL Classification: O14, O15, O41, J13, J23, O33, E24.

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1 Introduction

Modern economic growth began with industrialization. The British Industrial Revolution transformed a largely agrarian economy into one organized around manufacturing, mechanized production, and increasingly complex networks of suppliers, managers, clerks, engineers, accountants, and other producer-service occupations. In the vast majority of today's rich countries, the long-run movement of labor out of agriculture was accompanied by a substantial manufacturing phase before services came to dominate employment. Yet this path has become increasingly uncommon. Many developing countries now move directly from agriculture into services, reaching lower manufacturing employment peaks and deindustrializing at lower income levels than earlier industrializers (Rodrik, 2016).

Can countries develop without manufacturing? A narrow answer is yes: labor can leave agriculture and move into services. But this paper argues that the path out of agriculture matters for the kind of service economy that emerges. Manufacturing is not only a sector that produces goods. It also creates demand for high-skill producer services: accounting, management, technical supervision, engineering, logistics, purchasing, quality control, and other coordination tasks. These tasks organize production and generate transferable routines, standards, management systems, and supplier-coordination capabilities. I call this stock of transferable routines and know-how *codified organizational knowledge*. Once created, it can become the foundation for a high-skill service economy.² By contrast, economies that bypass manufacturing may enter services earlier, but with employment concentrated in lower-skill consumer-facing activities such as retail, restaurants, hospitality, and personal services. Manufacturing therefore does not just delay the rise of services; it changes the composition of service work, the demand for skills, and in so doing also the education and fertility choices made during the transition out of agriculture.

I provide evidence for this mechanism in two steps. First, I document cross-country facts linking the size of the manufacturing phase to later development outcomes. Countries with higher peak manufacturing employment shares grow faster after they have largely transitioned into services, accumulate more tertiary education, experience faster fertility declines, and develop more skill intensive service sectors. These patterns hold among economies with high service employment shares, suggesting that manufacturing matters not only because it absorbs labor during the transition, but because it shapes the service economy that follows. Second, I use full count U.S. Census data from 1850 to 1940 to study the mechanism within a single economy. Counties with

2. A historical example is the diffusion of factory-born management methods into services. Scientific management were developed to make repeated industrial work measurable and reorganizable; Lillian Gilbreth later applied these methods to department-store work at Macy's and to other service settings. Similarly, management accounting and budgeting tools developed for complex industrial and procurement problems later became portable tools for retail and service firms through the work of John McKinsey.

larger manufacturing booms later developed service occupations with higher educational content, measured using occupation-level education scores from the 1950 Census. The effects appear after the manufacturing boom, are strongest for counties with the largest manufacturing peaks, and persist for over a century until the last available data in the early 2000s. IV estimates based on geography-driven manufacturing exposure point to the same conclusion.

Building on this evidence, I build a unified growth model with structural change, fertility, education, and heterogeneous producer services. In the model, manufacturing raises demand for organizational and technical producer-service tasks that are intensive in skilled labor. This raises the return to education and builds codified organizational knowledge that persists after manufacturing declines. Economies with larger manufacturing phases therefore enter the service economy with more human capital, lower fertility, and a more skill intensive producer-service sector. Calibrated to the U.S. transition from 1850 to 2019, the model matches the decline of agriculture, the manufacturing hump, the rise of services, the fertility transition, and the high-skill content of producer-service inputs. Counterfactual economies with smaller manufacturing peaks reach lower income, grow more slowly, accumulate less education, have higher population growth, and develop less skill intensive services. Quantitatively, the mechanism explains 31 percent of cross-country log GDP per capita gaps, 60 percent of GDP per capita growth differences, 53 percent of tertiary education differences, and 39 percent of cumulative population growth differences.

This paper contributes to three literatures. First, it contributes to work on structural transformation and premature deindustrialization. Classic and modern theories of structural change study why labor moves from agriculture to manufacturing and services, emphasizing non-homothetic preferences, sectoral productivity differences, and relative prices (Kuznets, 1966, 1973; Lewis, 1954; Rostow, 1959; Matsuyama, 1992; Kongsamut et al., 2001; Ngai and Pissarides, 2007; Herrendorf et al., 2014; Boppart, 2014; Comin et al., 2021). Recent work studies why manufacturing peaks differ across countries, emphasizing premature deindustrialization, trade, global specialization, technology gaps, and heterogeneous sectoral productivity growth (Rodrik, 2016; Matsuyama, 2009; Uy et al., 2013; Sposi et al., 2021; Huneus and Rogerson, 2024; Fujiwara and Matsuyama, 2024). I focus instead on the consequences of these differences. I ask whether missing the manufacturing phase changes the development path itself by altering skill demand, fertility choices, knowledge accumulation, and the long-run composition of services.

Second, the paper contributes to work on service led growth and the organization of production. A growing literature asks whether services can substitute for manufacturing as an engine of development and emphasizes that services are highly heterogeneous (Fan et al., 2023; Chen et al., 2023; Peters et al., 2026; Herrendorf et al., 2022; Buera and Kaboski, 2012a,b; Buera et al., 2022; Duernecker and Herrendorf, 2022; Duernecker et al., 2024). Recent work argues that, as manufacturing becomes less

labor-absorbing, developing countries must find ways to raise productivity in labor-absorbing services (Rodrik and Sandhu, 2025; Rodrik and Stiglitz, 2025). This paper asks a complementary question: why do some economies enter the service era with the skills and organizational capabilities needed for such upgrading, while others do not? Recent work further shows that modern services can generate productivity growth when firms are able to scale operations across markets and when the economy has sufficient human capital and managerial capacity (Hsieh and Rossi-Hansberg, 2023). Other work shows that development involves the reorganization of production toward larger, more skill intensive firms and white-collar tasks (Engbom et al., 2025). My contribution is to connect this organizational margin to the sectoral path of development. Manufacturing matters not simply because it has high productivity growth, but because it helps determine whether the service economy expands into high-skill producer services or into low-skill consumer-facing services. This mechanism also connects the paper to work on urbanization without industrialization and consumption cities, where labor leaves agriculture without necessarily entering modern manufacturing (Gollin et al., 2016; Jedwab et al., 2022).

Third, the paper contributes to unified growth theory. Canonical unified growth models explain how technological progress, human capital accumulation, and fertility decline interact in the transition from stagnation to modern growth (Galor and Weil, 2000; Hansen and Prescott, 2002; Galor, 2005, 2011; Becker et al., 1990; Doepke, 2004; Galor and Moav, 2004). I embed this transition in a model of structural change across agriculture, manufacturing, and services. This matters because sectors differ in their demand for skills and in their contribution to codified organizational knowledge. The path out of agriculture therefore affects not only sectoral employment shares, but also fertility, education, and the organizational frontier inherited by the service economy. In this sense, the paper links two literatures that have often been studied separately: structural change across sectors and the joint evolution of fertility, human capital, and income.

The rest of the paper is structured as follows. Section 2 documents the cross-country facts relating peak manufacturing employment to GDP per capita growth, education, service sector composition, and fertility. Section 3 uses full count U.S. Census data to study the effect of historical manufacturing booms on the skill content of services across counties. Section 4 develops the model. Section 5 describes the calibration to the United States over 1850-2019. Section 6 presents the counterfactual results. Section 7 concludes. Appendix A describes the data construction, Appendix B reports additional cross-country evidence, Appendix C reports additional U.S. county evidence, Appendix D contains model details, Appendix E reports calibration details and robustness, and Appendix F describes policy extensions.

2 Cross-Country Evidence

This section documents three cross-country facts linking the size of the manufacturing phase to later development outcomes. Countries with larger manufacturing employment peaks grow faster after moving into services, accumulate more tertiary education, develop more skill intensive service sectors, and experience faster fertility declines. The analysis combines data on GDP per capita, fertility, population, and broad sectoral employment shares from Gapminder with longer sectoral series from the GGDC 10-Sector Database, tertiary education measures from [Barro and Lee \(2013\)](#), service sector microdata from IPUMS International, and input-output linkages from the OECD Inter-Country Input-Output tables. The baseline sample includes 174 countries with employment shares over 1990–2018 and long-run GDP per capita, population, and fertility data. Appendix A describes the data sources, sample construction, and variable definitions in detail.

FACT 1: *Countries with higher peak manufacturing employment shares grow faster as they transition into services.*

The rise of services is often viewed as a drag on aggregate growth. In the standard Baumol cost-disease logic, services experience slower productivity growth than agriculture or manufacturing and as labor reallocates toward them, aggregate productivity growth declines ([Baumol, 1967](#); [Duernecker and Sanchez-Martinez, 2023](#); [Duernecker et al., 2024](#)). However, little attention has been given to whether this pattern is different depending on how service economies reach this state, namely whether they went through manufacturing or not.

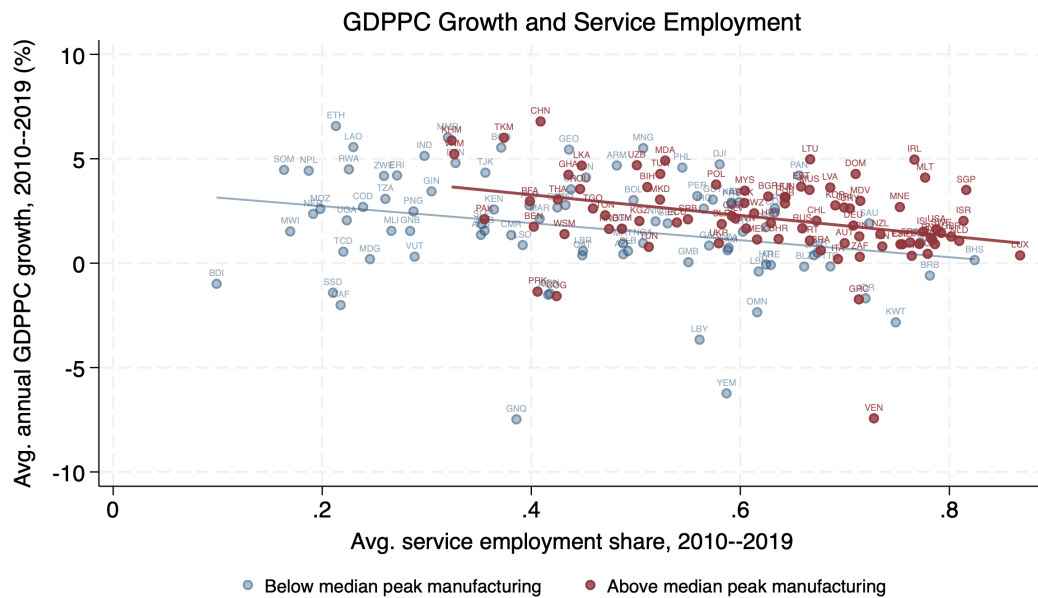


Figure 1: Baumol’s cost disease. Countries with above median peak manufacturing employment shares continue to have higher growth rates as they move into services. Data source: ILO and World Bank via Gapminder.

Figure 1 plots the average yearly growth rate of PPP real GDP per capita over 2010-2019 across 174 countries against their average service sector employment share in the same period, distinguishing between countries with above and below median peak manufacturing employment share, which is the maximum employment share in manufacturing over the sample period.³ While across countries a higher services employment share is associated with a lower aggregate growth rate, those that reached a peak manufacturing employment share above the median of the cross country distribution grow faster at all levels of the services employment share. To get a sense of the magnitude of this relationship I estimate the following simple regression model:

$$y_c = \alpha_r + \alpha_i + \Gamma' X_c + \beta^s \text{Serv. Emp. Share}_c + \beta^m \text{Manu. Peak}_c + \epsilon_c$$

Where y_c is the average growth rate of GDP per capita over 2010-2019, α_i and α_r are, respectively, income quintile and regional fixed effects using the five World Bank region definitions, X_c is a vector of control variables including the institutional quality index from the World Bank's ease of doing business dataset, average GDP per capita over 2010-2019 and years since the peak of manufacturing employment. The results are

Table 1: Dep. Var. is Avg GDPPC growth rate 2010-2019

	(1)	(2)	(3)	(4)
Avg. Serv. Emp. Share, 2010–2019	-2.742*** (-2.84)	-5.171*** (-4.49)	-7.566*** (-3.59)	-9.962*** (-4.17)
Peak Emp. % Man.		9.658*** (4.59)	5.455** (2.16)	5.511** (2.09)
N	174	174	174	174
R2	0.0469	0.130	0.215	0.284
Income Quintile FE	no	no	yes	yes
Region FE	no	no	yes	yes
Controls	no	no	no	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

reported in Table 1. The coefficient on peak manufacturing employment is positive and statistically significant across specifications, and its magnitude is economically meaningful. In the most saturated specification, a one percentage point higher manufacturing peak is associated with an offset of more than half of the growth slowdown associated with a one percentage point higher service employment share. As shown below, this relationship is consistent with countries with larger manufacturing peaks accumulating more human capital and developing more skill intensive services.

3. I exclude 2020 due to the effect of the COVID-19 pandemic on output. The peak manufacturing employment share is the maximum observed employment share in manufacturing over 1990-2018 for the countries in the ILOEST data and 1960-2018 for those also appearing in the GGDC 10-Sector database.

FACT 2: Countries with higher peak manufacturing employment shares accumulate more tertiary education and develop more skill intensive services.

The growth patterns in Fact 1 are accompanied by differences in human capital accumulation. If manufacturing intensive development is associated with greater demand for skill intensive tasks, then countries with larger manufacturing phases should also display larger increases in education and a more skill intensive service sector. Figure 2 plots tertiary education against GDP per capita, separately for countries above and below the median peak manufacturing employment share. Tertiary education rises with income in both groups, but the relationship is steeper among countries with larger manufacturing peaks. This cross-sectional pattern also appears within countries over time. To summarize the relationship, I estimate

$$\Delta e_c = \alpha + \gamma \Delta \log y_c + \sum_{q=2}^5 \beta_q \mathbb{1} \{ \text{Manu. Peak}_c \in Q_q \} + \varepsilon_c.$$

Here $\Delta e_c = e_{c,T_c} - e_{c,t_c}$ is the change in tertiary education for country c between the first and last years in which both education and GDP per capita are observed over 1960–2020, and $\Delta \log y_c = \log y_{c,T_c} - \log y_{c,t_c}$ is the corresponding change in log GDP per capita. The omitted category is the lowest peak-manufacturing quintile, so the coefficients β_q compare education gains in higher peak-manufacturing quintiles to those in the lowest quintile, conditional on income growth.

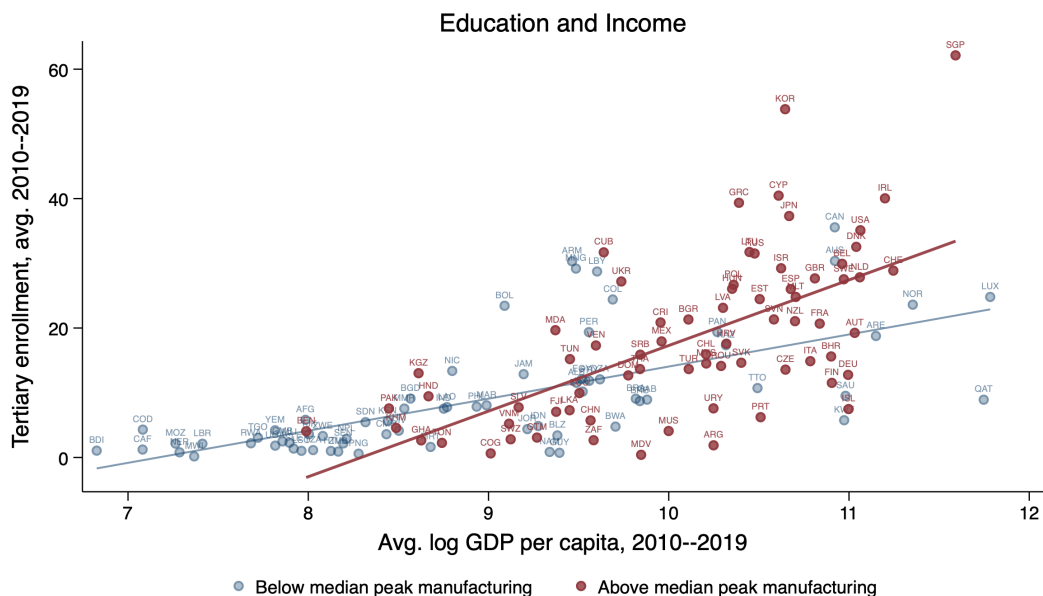


Figure 2: Tertiary education and GDP per capita. Tertiary education rises with development, but more steeply among countries with above-median peak manufacturing employment. Data source: Barro and Lee (2015) and ILO via Gapminder.

Table 2: Education Gains by Peak Manufacturing Quintile

	Change in education
$\Delta \log$ GDP per capita	5.391** (2.215)
Peak manufacturing Q2	5.337** (2.160)
Peak manufacturing Q3	4.184** (1.919)
Peak manufacturing Q4	7.647*** (1.979)
Peak manufacturing Q5	12.300*** (2.315)
Constant	-0.653 (1.049)
Observations	139
R-squared	0.296

Notes: Robust standard errors in parentheses. The omitted category is Q1, the lowest peak-manufacturing quintile. The dependent variable is the change in education between the first and last usable country-year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2 reports the results. Conditional on income growth, countries with larger manufacturing peaks experience larger increases in tertiary education. In the highest peak-manufacturing quintile, the increase is up to 12.3 percentage points larger than in the lowest quintile. I then ask whether this education pattern is mirrored in the composition of services. Using the 2015 OECD Inter-Country Input-Output table, I compute the service sector output induced by a \$100 final-demand increase in agriculture, manufacturing, or consumer services:

$$x_j = (I - A)^{-1} f_j,$$

where A is the matrix of technical input coefficients and f_j is the sector-specific final-demand shock. I classify finance and insurance, computer and information services, and professional, scientific, and technical activities as high-skill services, and the remaining service industries as low-skill services. For the definition of consumer services see Appendix B.

Figure 3 reports the resulting service sector mix. The service output induced by manufacturing contains a larger high-skill component than the service output induced by consumer-service demand. This pattern is consistent with the view that manufacturing is linked to a more skill intensive part of the service sector, not only through manufacturing employment itself, but also through its intermediate-input linkages. Agriculture also induces some high-skill service activity, but agricultural expenditure shares decline with development (Herrendorf et al., 2013). The input-output evidence therefore suggests that, among the sectors expanding during the transition out of agriculture, manufacturing is more closely connected to the high-skill service activities

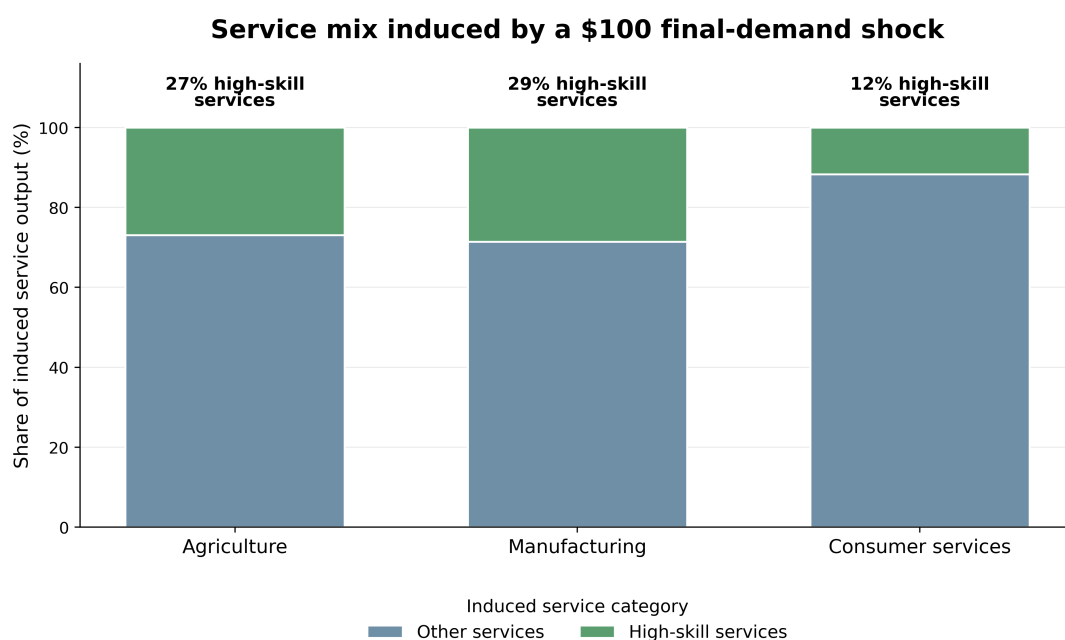


Figure 3: service sector output induced by a \$100 final-demand increase in agriculture, manufacturing, or consumer services, computed using Leontief total requirements. Consumer services are industries where final consumption exceeds 50 percent of gross output. Data source: 2015 OECD ICIO.

measured here. Appendix B shows that the pattern is similar across country income groups and under broader definitions of consumer-service demand. The education and input-output patterns are consistent with higher skill demand in economies with larger manufacturing phases. I next examine the demographic counterpart of this pattern. In standard quantity-quality models, higher returns to education induce households to invest more in child quality and reduce fertility (Becker et al., 1990; Galor and Weil, 2000; Galor, 2005, 2011). If larger manufacturing phases are associated with higher returns to education, they should therefore also be associated with faster fertility declines.

***FACT 3:** Countries with higher peak manufacturing employment shares experience faster demographic transitions, exhibiting lower fertility at comparable levels of development.*

Figure 4 plots the mean fertility rate over the log of GDP per capita in countries in the above median and below median peak manufacturing employment share groups, using data on fertility and GDP per capita using a balanced cross-country panel back to 1800. Countries with higher peaks had lower fertility rates even at relatively low levels of GDP per capita, for comparison purposes a log GDP per capita of 6.5 corresponds to that of a median low income country today. This implies that even early in the development process, when today's industrialized countries had relatively low income levels, they still had lower fertility levels than a non industrializing country did at a similar level of development. To formalize this analysis, I run the following descriptive regression:

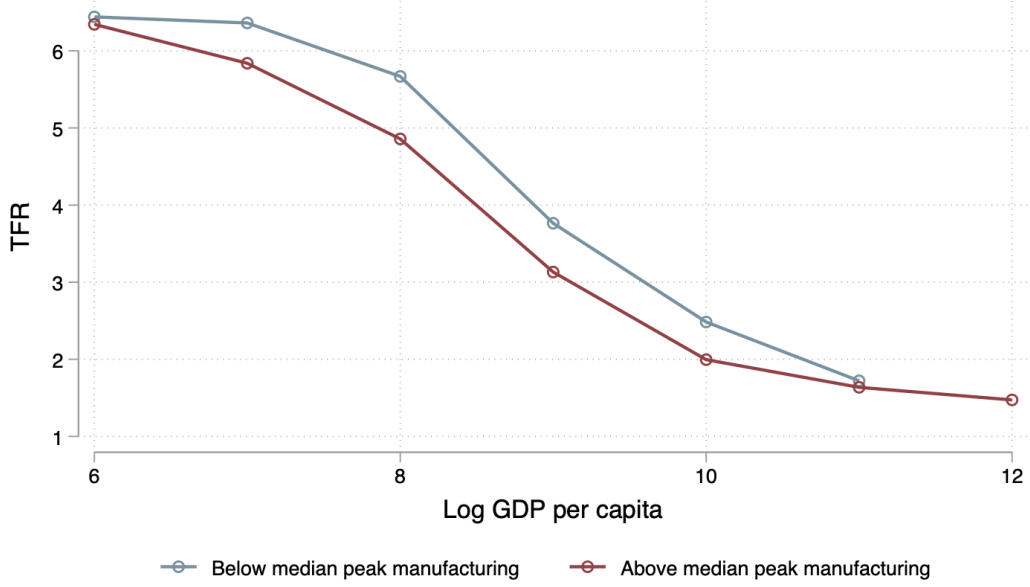


Figure 4: Scatterplot of the Total Fertility Rate (TFR) over log of GDP per capita across 176 countries observed from 1800 to 2020. Countries are grouped by being above and below median peak manufacturing employment share. Data Source: Gapminder.

$$\begin{aligned} \overline{TFR}_c = & \alpha + \beta \text{Manu. Peak}_c + \gamma_1 TFR_{c,1800} + \gamma_2 \log GDPpc_{c,1800} \\ & + \gamma_3 \text{Peak Year}_c + \delta_r + \varepsilon_c. \end{aligned}$$

The dependent variable, $\overline{TFR}_c$, is the average fertility rate observed for country c over 1800-2020 and Manu. Peak_c is the country's peak manufacturing employment share. I control for initial fertility, initial log GDP per capita, and the year the country reached its manufacturing peak and regional fixed effects δ_r .

The results are reported in Table 3. Across specifications, the coefficient on peak manufacturing is negative and statistically significant. This relationship is not simply a comparison between rich and poor countries: it remains after controlling for initial income and initial fertility, and after adding region fixed effects. Together with Figure 4, the evidence suggests that countries that ultimately reached higher manufacturing employment peaks tended to have lower fertility at comparable stages of income development. Taken together, the three facts point to a common pattern: economies with larger manufacturing phases grow faster, accumulate more human capital, develop more skill intensive services, and experience faster fertility declines. These cross-country relationships are descriptive, but they motivate the mechanism studied in the rest of the paper. The next section turns to U.S. county evidence, where manufacturing booms can be measured at a more disaggregated level and followed over time.

Table 3: Dep. Var. is Average Fertility Rate 1800-2020

	(1)	(2)	(3)	(4)
Peak Emp. % Man.	-1.788* (-1.80)	-1.036*** (-2.75)	-1.184*** (-3.81)	-0.700*** (-2.82)
Fertility rate 1800		0.598*** (13.97)	0.387*** (8.70)	0.316*** (5.80)
log GDP per capita 1800			-0.810*** (-5.54)	-0.574*** (-4.28)
Year of manufacturing peak			0.00701* (1.75)	0.00451 (1.28)
N	189	189	176	176
R2	0.0465	0.513	0.642	0.760
Region FE	no	no	no	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

3 U.S. County Evidence

The cross-country evidence in Section 2 suggests that economies with larger manufacturing phases later develop more skill-intensive service sectors, accumulate more human capital, and experience faster demographic transitions. This section studies the same relationship within the United States during the main period of American industrialization. The county setting holds fixed national institutions, aggregate shocks, and measurement systems, while allowing me to observe the timing and size of local manufacturing booms. I ask whether counties with larger manufacturing peaks subsequently developed more education-intensive service sectors, whether this upgrading is concentrated in occupations connected to business organization and intermediation, and whether historical manufacturing peaks predict persistent differences in local development.

I measure county-level manufacturing exposure using county tabulations from the Census of Manufactures over 1880–1940 compiled by [Haines \(2001\)](#). Since consistent county employment denominators are not available before 1950, I define a county’s manufacturing peak as the maximum manufacturing employment-to-population ratio observed over this period. Earlier manufacturing censuses are omitted because of well-known reliability issues ([Vickers and Ziebarth, 2024](#)). I combine these manufacturing data with the full-count U.S. Censuses from 1850 to 1940, restricting the sample to fathers of respondents aged 6 to 20 with non-missing industry and employment information.⁴ The main outcome is the skill content of service employment. Because

4. I use a common sample to compute county-by-year enrollment rates, education scores, and fertility rates. Appendix A describes the data construction in detail.

individual education is not observed before 1950, I use an occupation-based education score: for each three-digit occupation, IPUMS assigns the share of workers in that occupation in 1950 who had completed some higher education. I then average this score across workers employed in service industries in each county-year. The resulting variable, $EDSCOR50_{ct}$, captures changes in the occupational composition of services toward occupations that were education-intensive in 1950.

I estimate event-study effects around each county's manufacturing peak. Let T_c^{peak} denote the decade in which county c reaches its maximum manufacturing employment-to-population ratio. For each peak-manufacturing group $\mathcal{P}$, I estimate dynamic effects of the form

$$EDSCOR50_{ct} = \delta_c + \gamma_t + \sum_k \beta_k \mathbf{1}\{t - T_c^{peak} = k\} \times \mathbf{1}\{\text{Manu. Peak}_c \in \mathcal{P}\} + \varepsilon_{ct}, \quad (1)$$

where δ_c are county fixed effects and γ_t are census-decade fixed effects. I implement the event study using the imputation estimator of [Borusyak et al. \(2024\)](#), which constructs untreated potential outcomes from not-yet-treated and untreated observations and then averages treatment residuals by event time. The coefficients trace whether counties in a given part of the manufacturing-peak distribution experience differential changes in service-sector education scores before and after their manufacturing peak.

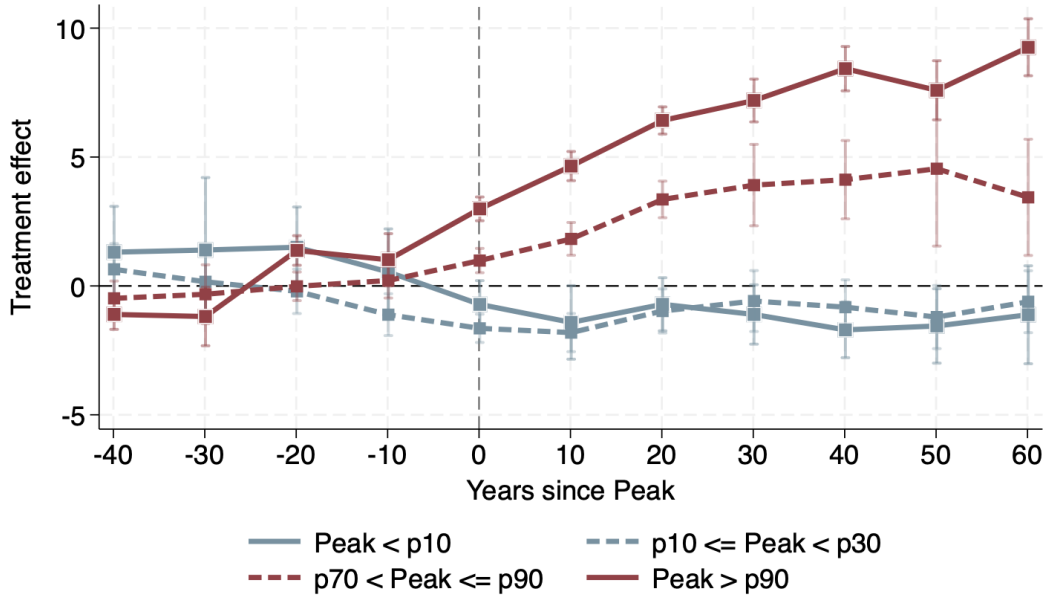


Figure 5: Event-study estimates by mutually exclusive peak-manufacturing bins. The outcome is the mean 1950 occupation education score among service-sector workers in county c and decade t . Counties are grouped by their position in the cross-county distribution of Manu. Peak_c . Event time is measured in years relative to the county manufacturing peak. Estimates are obtained using the Borusyak, Jaravel, and Spiess imputation estimator with county and census-decade fixed effects.

Figure 5 shows that counties in the top decile of the manufacturing-peak distribution experience a clear increase in the education score of service occupations after their peak. The pre-peak estimates are close to zero, while the post-peak estimates rise and remain positive several decades after the peak. The pattern indicates that larger manufacturing peaks are followed by a divergence in the skill intensity of local service employment. Appendix C.3 shows that the result is robust to using finer bins of the manufacturing-peak distribution. Appendix C.4 shows that these effects remain essentially unchanged after controlling for contemporaneous urban share and log county population, and that a geography-predicted manufacturing peak produces a similar event-study pattern.

I next examine which service occupations account for the post-peak increase. The aggregate service education score includes health, education, religion, and welfare occupations, which have consistently high education scores and are likely to respond strongly to income growth, population scale, public-service demand, and urban development. To isolate the component of service upgrading most directly related to market services, I decompose the effect after excluding these occupations. For each occupation group g , I construct its contribution to the service education score as

$$Contribution_{ct}^g = \frac{\sum_{i \in g} w_i EDSCOR50_i}{N_{ct}^{-HERW}}, \quad (2)$$

where N_{ct}^{-HERW} is total service employment excluding health, education, religion, and welfare occupations. The mutually exclusive groups are organizational/business services, technical services, consumer/personal services, and other services. Organizational/business services include managers, accountants, bookkeepers, clerks, secretaries, purchasing agents, legal and financial intermediaries, and related coordination workers. Figure 6 shows that the positive contribution is concentrated in organizational and business occupations. Technical services, including engineers, architects, chemists and others, contribute modestly, while consumer and residual service occupations do not contribute positively. Thus, the post-peak increase outside health, education, religion, and welfare is not a broad skill-upgrading across all remaining service occupations. It is driven by increases in occupations connected to firm organization, administration, coordination, finance, law, and business intermediation. Appendix C.5 performs this decomposition for individual occupations, including health, education and religion, reaching a similar conclusion.

I finally ask whether historical manufacturing peaks are associated with persistent differences in local development. Using county outcomes measured in 2000, I estimate

$$y_{c,s,2000} = \delta_s + \beta \text{Manu. Peak}_{c,s} + \Gamma' X_{c,s} + \epsilon_{c,s}. \quad (3)$$

Here $y_{c,s,2000}$ is a county outcome measured in 2000 and δ_s is a state fixed effect. The variable $\text{Manu. Peak}_{c,s}$ is the county's historical manufacturing peak. The vector $X_{c,s}$ includes baseline historical controls measured in 1850: log population, urbanization,

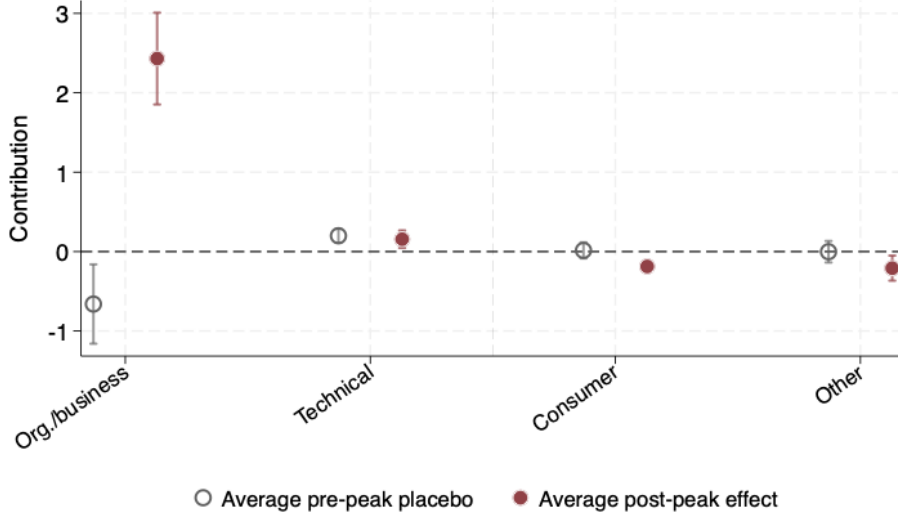


Figure 6: Occupational decomposition of the post-peak effect among service occupations excluding health, education and religion. Each point reports the average event-study effect for a mutually exclusive occupation group. The outcome is the group’s contribution to the service education score. Red markers report the average post-peak effect; gray markers report the average pre-peak placebo.

school enrollment, and the number of colleges. To address the concern that realized manufacturing peaks are endogenous to other county characteristics, I also construct a geography based predicted manufacturing peak. The prediction uses the fact that the geographic determinants of manufacturing changed over time: proximity to the fall line was especially important for early industrialization because it provided access to water power, while proximity to coal became more important later. I estimate

$$\widehat{\text{Manu. Peak}}_c = \sum_d \left(\hat{\beta}_d^C \ln(\text{coal}_c) + \hat{\beta}_d^F \ln(\text{fall}_c) + \hat{\beta}_d^O \ln(\text{ohio}_c) + \hat{\beta}_d^{GL} \text{GreatLakes}_c \right) \cdot \mathbf{1}\{\text{peak decade} = d\}. \quad (4)$$

The geographic variables are distance to the nearest coal mine active in the late nineteenth century, distance to the Ohio River, distance to the fall line, and an indicator for adjacency to the Great Lakes.⁵ I use this geography-based prediction as an instrument for the realized manufacturing peak in the 2000 cross-sectional specifications.

Table 4 reports the results. Counties with larger historical manufacturing peaks have higher patenting, higher college attainment, higher income, larger innovation employment shares, and more high-skill service employment in 2000. Panel B instruments the realized manufacturing peak using the geography-based predicted peak in equation (4). The IV estimates are generally larger than the OLS estimates, though less precise for some outcomes. Appendix A describes the construction of the 2000 outcomes, including patents per capita, innovation employment, high-skill services, and

5. Data sources are described in Appendix A.

Table 4: Effect of Peak Manufacturing Employment Share on Outcomes in 2000

	Log patents pc	Frac. college+	Log income	Innovation / total emp.	HS services / total services	Advanced mfg / total mfg
Panel A: OLS						
Peak Emp. % Man.	0.031*** (0.006)	0.127*** (0.036)	0.488*** (0.105)	0.055*** (0.021)	0.156*** (0.052)	0.080* (0.042)
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Historical Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1600	1600	1600	997	997	997
R^2	0.192	0.317	0.401	0.226	0.133	0.300
Panel B: IV						
Peak Emp. % Man.	0.042** (0.018)	0.421*** (0.156)	0.774 (0.471)	0.190* (0.099)	0.533* (0.283)	0.038 (0.125)
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Historical Controls	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen–Paap	83.91	83.91	83.91	83.91	83.91	83.91
Observations	1600	1600	1600	997	997	997
R^2	0.189	0.285	0.396	0.181	0.096	0.299

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

advanced manufacturing. Additional appendix evidence supports the same interpretation: Appendix C.2 shows that higher lagged manufacturing employment shares predict fertility declines, school enrollment increases, and increases in the service-occupation education score within counties over 1850–1940; Appendix C.7 shows that counties with larger manufacturing peaks account for a larger share of university openings over 1800–2000; and Appendix C.8 shows that new manufacturing establishments are followed by increased help-wanted advertisements for skilled manufacturing and skilled service occupations, including technical, administrative, and managerial roles.

Overall, the U.S. evidence mirrors the cross-country patterns. Larger manufacturing booms are followed by more education-intensive service sectors; the occupational decomposition shows that the non-health, non-education component of this upgrading is concentrated in organizational and business service occupations; and counties with larger historical manufacturing peaks exhibit persistent differences in innovation, education, income, and high-skill service employment in 2000. The next section interprets this evidence through the lens of a growth model linking manufacturing to skill demand, education, fertility, and service composition.

4 Model

In this section I build a simple model that can speak to these facts and to the broader question of how manufacturing shapes the development process. The model is organized around one key friction: sectors differ in the codifiability of the producer-service tasks they generate. This is natural because different sectors expose the economy to different organizational problems, some of which are easier to standardize, transfer, and reorganize around skilled labor. As a result, the size of the manufacturing phase can affect not only employment during the transition, but also the kind of service economy that emerges afterward. The rest of the model adds only the ingredients needed to trace this mechanism over time. It distinguishes consumer services from producer services, because the question is not only whether services expand, but what kind of service economy is created in the transition. It allows education and fertility to respond endogenously, because manufacturing intensive development is associated with both human capital accumulation and a quicker demographic transition. And it contains a persistent organizational state, because the effects of manufacturing appear after manufacturing employment has declined.

The model implements this mechanism in a unified-growth environment with structural change, endogenous fertility and education, and producer-service tasks that differ in codification resistance across sectors. The rest of this section describes the environment, household choices, production, and dynamics. Section 5 takes the model to the U.S. transition, and Section 6 uses the calibrated economy for counterfactuals.

4.1 Environment

Time is discrete. The economy is populated by overlapping generations who live for two periods: childhood and adulthood. During childhood, individuals receive education and decide whether to become skilled. During adulthood, they work, consume, choose fertility, and invest in the education of their children.

There are three final-good sectors, indexed by $j \in \{a, m, c\}$: agriculture, manufacturing, and consumer services. Final goods are consumed by households and produced using sectoral value added, agricultural and manufacturing intermediates, and producer services. Producer services are intermediate inputs used by all final-good sectors. They can be performed in low-skill mode or high-skill mode. High-skill producer services correspond to the organizational, technical, professional, and managerial tasks that become more important as production becomes more complex.

The key friction is that sectors differ in how easily their producer-service tasks can be reorganized into high-skill form. In each final-good sector, producer-service tasks are indexed by a codification hurdle. A task is performed in high-skill mode when doing so is cheaper than performing it in low-skill mode. The set of high-skill tasks is therefore endogenous and depends on the relative price of skilled labor and on

codified organizational knowledge H_t . A higher H_t lowers the cost of reorganizing tasks into high-skill form.

Households are indexed by parental origin $i \in \{a, m, s\}$, corresponding to whether the adult generation works in agriculture, manufacturing, or services.⁶ Parents allocate time between market work, child rearing, and child education. Fertility determines population growth, education affects children's human capital, and the skill premium shapes the future supply of skilled labor.

At the end of each period, population, the skilled share, sectoral productivity, and codified organizational knowledge are updated. Children then become adults and make the same choices.

4.2 Households

The household block embeds the quantity-quality mechanism of unified growth theory in a multisector environment. As in [Galor and Weil \(2000\)](#), growth raises the return to human capital by making inherited knowledge less useful. I use the tractable human capital formulation in [Lagerlöf \(2006\)](#) to discipline this channel quantitatively.

Adults are indexed by the sector in which they work, $i \in \{a, m, s\}$. Sector i affects the fixed time cost of raising children, τ_i , which allows the model to match fertility differences across adults employed in different sectors. I otherwise impose a common wage and education environment across households. Let $w_{u,t}$ and $w_{s,t}$ denote unskilled and skilled wages, and let S_t denote the aggregate skilled share. The wage scale entering the household problem is

$$w_t(S_t) = (1 - S_t)w_{u,t} + S_t w_{s,t}. \quad (5)$$

The time cost of education is

$$\tau_e(S_t) = (1 - S_t)\tau_e^u + S_t \tau_e^s, \quad \tau_e^s < \tau_e^u. \quad (6)$$

Thus sectoral heterogeneity enters fertility through τ_i , while the wage and education environment are common across households. This aggregation keeps the household block tractable: fertility is chosen at the household level, while education is produced in a broader environment shaped by the aggregate skilled share.

A household employed in sector i chooses fertility $n_{i,t}$ and education investment per child $e_{i,t}$. Preferences are

$$U_{i,t} = (1 - \gamma) \log C_t + \gamma \log (n_{i,t} h_{i,t}), \quad (7)$$

where C_t is composite consumption and $h_{i,t}$ is the human capital of each child. Final consumption is allocated across goods using the non-homothetic CES demand system

6. I use $i \in \{a, m, s\}$ to index parental origin, where s denotes service sector origin, and $j \in \{a, m, c\}$ to index final-good sectors, where c denotes consumer services.

of [Comin et al. \(2021\)](#):

$$\sum_{j \in \{a, m, c\}} \omega_j^{\frac{1}{\sigma}} \left(\frac{C_{j,t}}{C_t^{\varepsilon_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1. \quad (8)$$

Here $C_{j,t}$ is consumption of final good j , ω_j are sectoral weights, σ is the elasticity of substitution across final goods, and ε_j governs sector-specific income effects. This demand system allows income growth and relative prices to reallocate final demand across agriculture, manufacturing, and consumer services.

Manufacturing is the numeraire. The household budget constraint is

$$p_{a,t}C_{a,t} + C_{m,t} + p_{c,t}C_{c,t} \leq [1 - n_{i,t}(\tau_i + \tau_e(S_t)e_{i,t})] w_t(S_t). \quad (9)$$

Each adult has one unit of time. Raising a child requires time τ_i , and education requires an additional time cost $\tau_e(S_t)e_{i,t}$ per child. Because preferences are Cobb-Douglas and the wage scale is common, each household devotes the same share of potential income to consumption. Sectoral origin affects fertility and education through τ_i , but not the composition of final consumption.

Child human capital is

$$h_{i,t} = \frac{e_{i,t} + \tau_h}{e_{i,t} + \tau_h + g_t}, \quad (10)$$

where g_t is aggregate productivity growth. This captures the Galor-Weil obsolescence channel: when growth is high, inherited knowledge depreciates more quickly, raising the value of education.

Let $\chi_t \equiv \tau_e(S_t)$. The Cobb-Douglas structure gives closed-form fertility and education choices. Fertility is

$$n_{i,t} = \frac{\gamma}{\tau_i + \chi_t e_{i,t}}. \quad (11)$$

The education choice is

$$e_{i,t}^* = \max \left\{ \sqrt{\frac{g_t(\tau_i - \chi_t \tau_h)}{\chi_t}} - \tau_h, 0 \right\}. \quad (12)$$

Appendix D derives these expressions. Equations (11) and (12) summarize the demographic channel: higher growth raises education investment; education raises the time cost of children; fertility falls. Sectoral differences in τ_i allow fertility to differ across adults employed in agriculture, manufacturing, and services, while the response of education to growth is governed by the common aggregate education environment.

Children choose whether to become skilled before entering adulthood. Let

$$\pi_t = \frac{w_{s,t}}{w_{u,t}} \quad (13)$$

denote the skill premium. The probability that a child becomes skilled is

$$p_{s,t} = \frac{1}{1 + \exp\left(-\frac{\beta_\pi \log \pi_t - c_0}{\sigma_{\text{skill}}}\right)}. \quad (14)$$

The parameter β_π governs the responsiveness of skill acquisition to the skill premium, c_0 is the average cost of becoming skilled, and σ_{skill} controls the dispersion of idiosyncratic costs. The skill choice problem is common across household sectors; sectoral employment affects fertility through τ_i , while the incentive to acquire skill is determined by the aggregate skill premium.

4.3 Production and producer services

The production side combines structural change with an input-output structure. Following [Herrendorf et al. \(2013\)](#), households consume final expenditure categories, while production is organized through value added and intermediate inputs. This distinction is important because services play two roles. Consumer services are final goods. Producer services are intermediate tasks used by all final-good sectors.

There are three final-good sectors $j \in \{a, m, c\}$: agriculture, manufacturing, and consumer services. Value added in sector j is produced using skilled and unskilled labor:

$$VA_{j,t} = A_{j,t} \left[\alpha_j (h_s L_{s,j,t})^{\frac{\sigma_L-1}{\sigma_L}} + (1 - \alpha_j) (h_u L_{u,j,t})^{\frac{\sigma_L-1}{\sigma_L}} \right]^{\frac{\sigma_L}{\sigma_L-1}}. \quad (15)$$

Here $A_{j,t}$ is sectoral value added productivity, $L_{s,j,t}$ and $L_{u,j,t}$ are skilled and unskilled labor, h_s and h_u are skill specific efficiency units, α_j governs value added skill intensity, and σ_L is the elasticity of substitution across skill types.

Gross output is

$$Y_{j,t} = A_{go,j,t} VA_{j,t}^{\theta_{j,t}} M_{a,j,t}^{\gamma_{a,j,t}} M_{m,j,t}^{\gamma_{m,j,t}} PS_{j,t}^{\gamma_{ps,j,t}}, \quad (16)$$

$$\theta_{j,t} + \gamma_{a,j,t} + \gamma_{m,j,t} + \gamma_{ps,j,t} = 1. \quad (17)$$

Here $M_{a,j,t}$ and $M_{m,j,t}$ are agricultural and manufacturing intermediates, and $PS_{j,t}$ is the producer-service composite used by sector j . The input-output shares vary over the transition. In the calibration, production begins in 1850 as a value added technology, $\theta_{j,1850} = 1$, and input-output coefficients then transition toward their observed modern values. This is motivated by evidence that changes in U.S. industry output

reflect changes in input-output coefficients as well as final demand (Feldman et al., 1987).

Producer services can be performed in low-skill or high-skill mode:

$$LS_t = A_{ls,t} h_u L_{u,ls,t}, \quad HS_t = A_{hs,t} h_s L_{s,hs,t}. \quad (18)$$

Low-skill producer services are tasks performed through routine execution, local knowledge, or informal supervision. High-skill producer services are the same broad functions after they have been reorganized into codified routines that can be delegated to skilled workers. They include professional, managerial, technical, accounting, engineering, planning, logistics, and coordination tasks.

The distinction is not that some functions exist only in low-skill form and others only in high-skill form. Rather, the same producer-service function can be performed in either mode. Work organization can be carried out by foreman judgment or by formal workflow design. Quality can be checked by eye or through testing protocols and defect records. Inventory can be managed by visual inspection or by stock-control systems. Accounting can record transactions or become budgeting, cost accounting, and managerial control. In each case, moving from low-skill to high-skill mode requires codifying the task: defining inputs and outputs, recording information, specifying procedures, and creating decision rules that skilled workers can use.

Manufacturing is a natural setting for this switch because many of its producer-service tasks are repeated, measurable, and costly to perform informally. Factory operations make delays, defects, inventories, worker motions, input shortages, and unit costs observable. This makes tasks such as workflow design, quality control, procurement, maintenance, budgeting, and personnel management easier to standardize. Once codified, however, these routines are portable. A historical example is motion study: repeated factory operations made work organization measurable, turning supervision and workflow design into high-skill managerial and engineering tasks. Lillian Gilbreth, one of the pioneers of motion study, later applied these factory-born methods to department-store work at Macy's, including hiring, training, employee handling, and clerical organization. Appendix D.4 provides additional examples of producer-service routines that were first codified in industrial settings and later adopted by service organizations, including quality control, inventory management, budgeting, and personnel systems.

I summarize these differences with a sector-specific codification friction. In each final-good sector j , producer-service tasks are indexed by a codification hurdle $z \geq 0$. The hurdle is distributed according to

$$F_j(z) = 1 - e^{-z/\kappa_j}. \quad (19)$$

The parameter κ_j is codification resistance. Lower κ_j means that the producer-service task environment associated with sector j has more tasks with low codification hurdles. Equivalently, for a given stock of codified organizational knowledge and a given

skill premium, a larger share of sector j 's producer-service tasks can be reorganized into high-skill mode.

Let H_t denote codified organizational knowledge: the economy-wide stock of transferable routines, standards, management systems, and producer-service know-how. Its effective depth is

$$\phi(H_t) = \frac{H_t^\eta}{H_t^\eta + H_{1/2}^\eta}. \quad (20)$$

A higher H_t makes it easier to reorganize a producer-service task into high-skill form. It captures the fact that firms do not need to reinvent workflow design, accounting systems, quality protocols, or inventory routines from scratch once these practices have been codified elsewhere in the economy.

A task with hurdle z is performed in high-skill mode when the high-skill cost is below the low-skill cost. Let

$$p_{hs,t} = \frac{w_{s,t}}{A_{hs,t}h_s}, \quad p_{ls,t} = \frac{w_{u,t}}{A_{ls,t}h_u} \quad (21)$$

denote the efficiency-unit costs of high-skill and low-skill producer services. The cut-off hurdle below which tasks are performed in high-skill mode is

$$z_{j,t}^* = \phi(H_t) \max \left\{ 1 + \log \left(\frac{p_{ls,t}}{p_{hs,t}} \right), 0 \right\}. \quad (22)$$

The cutoff is higher when codified organizational knowledge is deeper and when high-skill producer services are cheaper relative to low-skill producer services.

The high-skill task share in producer services used by sector j is therefore

$$\begin{aligned} x_{j,t} &= F_j(z_{j,t}^*) \\ &= 1 - \exp \left(- \frac{\phi(H_t) \max \left\{ 1 + \log \left(\frac{p_{ls,t}}{p_{hs,t}} \right), 0 \right\}}{\kappa_j} \right). \end{aligned} \quad (23)$$

Equation (23) is the core producer-service mechanism. High-skill producer-service use rises when skilled producer services become cheaper, when codified organizational knowledge expands, or when the sector's producer-service tasks have lower codification resistance. Thus κ_j is not an additional productivity shifter. It governs how easily the producer-service tasks generated by sector j can be transformed from informal low-skill execution into codified high-skill routines.

The producer-service composite used by sector j is

$$PS_{j,t} = \left[x_{j,t} HS_t^{\frac{\sigma_{ps}-1}{\sigma_{ps}}} + (1 - x_{j,t}) LS_t^{\frac{\sigma_{ps}-1}{\sigma_{ps}}} \right]^{\frac{\sigma_{ps}}{\sigma_{ps}-1}}. \quad (24)$$

Appendix D shows how this two-mode representation is derived from task-level aggregation. The implication is that sectors with lower κ_j use high-skill producer services more intensively for a given skill premium and stock of codified organizational knowledge.

4.4 Dynamics

The dynamic block closes the feedback loop. Sectoral composition affects codified organizational knowledge. Codified organizational knowledge affects high-skill producer-service demand. Skill demand affects education, fertility, and future skill supply. The resulting skill scale feeds back into productivity growth.

The aggregate state is

$$\Omega_t = \left(L_t, S_t, \{A_{j,t}\}_{j \in \{a,m,c\}}, H_t \right), \quad (25)$$

where L_t is population, S_t is the skilled share, $A_{j,t}$ are sectoral value added productivities, and H_t is codified organizational knowledge.

Population evolves according to aggregate fertility:

$$L_{t+1} = n_t L_t, \quad n_t = \sum_{i \in \{a,m,s\}} \ell_{i,t} n_{i,t}, \quad (26)$$

where $\ell_{i,t}$ is the employment share of adults of origin i . The skilled share evolves according to the skill choices of the young:

$$S_{t+1} = p_{s,t}. \quad (27)$$

The aggregate growth rate entering the household problem is the employment-share-weighted growth rate of sectoral value added TFP:

$$g_t = \sum_{j \in \{a,m,c\}} \ell_{j,t} \left(\frac{A_{j,t}}{A_{j,t-1}} - 1 \right). \quad (28)$$

Codified organizational knowledge is the stock of portable routines created when producer-service tasks are reorganized into high-skill form. Examples include workflow standards, quality-control protocols, inventory systems, budgeting routines, personnel practices, and supplier-management procedures. The state H_t denotes the economy-wide frontier of such transferable routines, production standards, management systems, and producer-service know-how. Education matters because identifying, formalizing, and transmitting tacit production practices is skill intensive, consistent with the Nelson–Phelps view that human capital facilitates technology adoption and diffusion (Nelson and Phelps, 1966; Benhabib and Spiegel, 1994) and with evidence linking management practices to the education of managers and workers (Bloom et al., 2012; Valero, 2021).

The same task-hurdle distribution that governs producer-service assignment disciplines the accumulation of H_t . Since

$$F_j(z) = 1 - \exp(-z/\kappa_j),$$

sector j 's codification hazard is

$$\lambda_j(z) = \frac{f_j(z)}{1 - F_j(z)} = \frac{1}{\kappa_j}.$$

Thus $1/\kappa_j$ measures the rate at which exposure to sector j 's producer-service task environment yields practices that can be standardized and transferred. With sectoral employment shares $s_{j,t}$, the economy-wide codification hazard is $\sum_j s_{j,t}/\kappa_j$. Letting $\bar{e}_t$ denote average education, the frontier supported by current education and sectoral exposure is

$$\tilde{H}_t = \bar{e}_t \sum_{j \in \{a,m,c\}} s_{j,t} \frac{1}{\kappa_j}. \quad (29)$$

Finally, codified organizational knowledge evolves as a frontier. Once routines and standards have been codified, they can be reused and diffused across firms and sectors. Hence

$$\begin{aligned} H_{t+1} &= \max\{H_t, \tilde{H}_t\} \\ &= H_t + \max\{\tilde{H}_t - H_t, 0\}. \end{aligned} \quad (30)$$

A temporary manufacturing phase can therefore have permanent effects whenever it supports a higher organizational frontier than the service economy would have generated on its own.

Finally, sectoral productivity evolves as

$$\frac{A_{j,t+1}}{A_{j,t}} = 1 + H_t (S_t L_t)^{\varphi_j}, \quad j \in \{a, m, c\}. \quad (31)$$

Productivity growth therefore depends on codified organizational knowledge and on the scale of skilled labor.

The mechanism can be summarized as follows. Structural change reallocates labor across sectors. Because sectors differ in codification resistance and skill intensity, this reallocation changes both the demand for skilled labor and the accumulation of codified organizational knowledge. Growth feeds back into household choices by raising the obsolescence of inherited knowledge, increasing education investment, and reducing fertility. Manufacturing affects long-run development when it leaves the service economy with a higher organizational frontier and a larger skilled share.

4.5 Limiting behavior and the service economy frontier

I now characterize the model's limiting behavior. Structural change makes GDP per capita growth non-stationary during the transition, so the relevant object is the asymptotic growth path, or AGP, reached once sectoral employment shares have converged (Buera et al., 2020). I focus on the service dominated branch,

$$\ell_{s,t} \rightarrow 1, \quad \ell_{a,t} \rightarrow 0, \quad \ell_{m,t} \rightarrow 0.$$

This is the empirically relevant limit for the question in this paper: once consumer and producer services absorb final-good employment, manufacturing can matter only through the states it leaves behind.⁷ Let H^* denote the organizational frontier inherited by the service economy.

Proposition 1 (Development path and terminal service AGP). *Fix primitives, including w_m . Under the regularity conditions in Appendix D.3, the model admits a unique development path from an agriculture-dominated low-growth region to a service dominated AGP. The terminal AGP is uniquely indexed by H^* . In particular, there exists a unique tuple*

$$(S^*, e^*, g^*, L^*)$$

such that

$$\begin{aligned} S^* &= S(H^*), \quad S'(H) > 0, \\ n_t &\rightarrow 1, \quad e_t \rightarrow e^* = \frac{\gamma - \tau_s}{\tau_e(S^*)}, \\ g^* &= \frac{(\gamma - \tau_s + \tau_e(S^*)\tau_h)^2}{\tau_e(S^*) (\tau_s - \tau_e(S^*)\tau_h)}, \end{aligned} \tag{32}$$

and

$$L^* = \frac{1}{S^*} \left(\frac{g^*}{H^*} \right)^{1/\varphi_c}. \tag{33}$$

Proof. See Appendix D.3.

The transition determines the inherited organizational frontier H^* . The inherited frontier determines S^* . The skilled share S^* determines g^* . Population scale then adjusts to satisfy

$$g^* = H^* (S^* L^*)^{\varphi_c}.$$

7. In the quantitative model, limiting employment shares depend on preferences, relative productivity, prices, and input-output linkages. The results below characterize the service dominated asymptotic branch. This is the empirically relevant case for modern high-income economies, where services account for most employment, and is consistent with structural-change models that characterize asymptotic growth after sectoral reallocation has converged; see Comin et al. (2021) and Buera et al. (2020).

Thus long-run growth is independent of population scale. In [Romer \(1990\)](#), larger research scale raises growth, generating the counterfactual implication that larger economies should grow faster. Semi-endogenous models remove this level scale effect, but long-run growth remains tied to population growth or research-effort growth ([Jones, 1995](#); [Kortum, 1997](#); [Segerstrom, 1998](#)). In a recent survey, [Jones \(2022\)](#) shows that this class of models cannot sustain growth in output per capita once population growth stops, something which is increasingly common in developed countries. In contrast, the model in this paper delivers positive growth, replacement fertility, and finite population. The reason is different from other no-scale models: in [Peretto \(1998\)](#), [Laincz and Peretto \(2006\)](#), and related work, scale effects disappear because entry and product variety adjust with market size; here, long-run growth is selected by the skill content of services, and the skill content of services is selected by the organizational frontier inherited from structural change.

I next characterize when manufacturing raises that frontier. To gain intuition consider the branching experiment in Figure 7. There are three periods. In each period, one sector absorbs all labor. Both economies start in agriculture and end in services. Economy *A* passes through manufacturing; economy *B* moves directly into services.

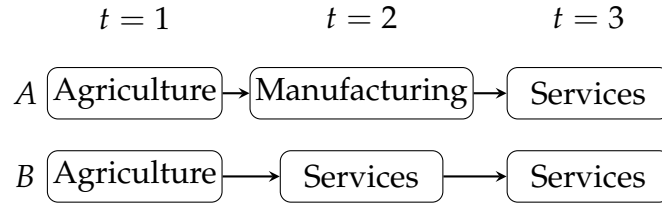


Figure 7: Canonical branching experiment.

Proposition 2 (Manufacturing versus direct services). *Consider the transition in Figure 7. Under the regularity conditions in Appendix D.3, and in the limiting case where $\tau_h \rightarrow 0$,*

$$g_A^* > g_B^* \iff H_A^* > H_B^* \iff \underbrace{(S_1 L_1)^{\varphi_m - \varphi_c}}_{\text{growth}} \underbrace{\frac{\tau_m}{\tau_s}}_{\text{quantity-quality}} > \underbrace{\left(\frac{\kappa_m}{\kappa_c}\right)^2}_{\text{codification}}.$$

Proof. See Appendix D.3.

The condition says that manufacturing raises long-run growth only if it raises the inherited organizational frontier H^* . It can do so in three ways. First, if $\varphi_m > \varphi_c$, manufacturing raises transition growth for a given $S_1 L_1$; higher growth raises education and therefore codified organizational knowledge. Second, if $\tau_m > \tau_s$, manufacturing origin households face higher fertility costs and choose more child quality, again raising education and H . Third, if $\kappa_m < \kappa_c$, a unit of education generates more codified organizational knowledge in manufacturing than in consumer services.

Evidence that manufacturing households are similar to service households in education and fertility therefore disciplines τ_m/τ_s , but it does not rule out a role for manufacturing in development.⁸ Equivalently, for the two paths in Figure 7 to converge to the same AGP, the direct-service path must compensate for any manufacturing advantage in transition growth or codification. If τ_m/τ_s is close to one, meaning that manufacturing and services households make the same choices, equalization requires either φ_c to be sufficiently high relative to φ_m , or κ_c to be sufficiently low relative to κ_m . Neither case is supported by the empirical evidence presented in this paper.

Appendix D.3 extends Proposition 2 to arbitrary transition paths with interior sectoral employment shares. In general, manufacturing raises long-run service growth if and only if it raises the organizational frontier inherited by services. Services can absorb labor, but an economy that bypasses manufacturing may enter consumer services with a lower organizational frontier, a lower skilled share, and a lower asymptotic growth rate.

5 Calibration

I calibrate the model to the United States over 1850–2019. The U.S. transition provides the benchmark economy for the counterfactual exercises in Section 6: over this period the United States moves out of agriculture, experiences a pronounced manufacturing hump, shifts into services, and completes most of its demographic transition. The calibration is designed to discipline both this aggregate development path and the mechanism emphasized in the paper. The aggregate moments discipline structural change, fertility, and income growth. The mechanism moments discipline the skill content of production and, in particular, the fact that the producer-service occupations used by manufacturing are more education-intensive than the producer-service occupations used by agriculture or consumer services.

A central issue in the calibration is how to define consumer services. In the baseline, I use a strict industry classification. I classify industries into agriculture, manufacturing, consumer services, and producer services, and treat consumer services as the subset of service industries whose output is primarily purchased by final consumers. This choice is conservative for two reasons. First, countries differ substantially in the extent to which final consumers purchase services that are also heavily used by firms, such as finance, professional services, and information services. The same is likely true over the long U.S. transition. A final-expenditure-weighted definition based on modern input-output tables may therefore misclassify the service bundle relevant for earlier periods, when the household consumption of these mixed-use services was plausibly very different. The strict classification instead isolates service industries that are likely

8. [Peters et al. \(2026\)](#) show that manufacturing households are not systematically different from households in consumer services along several observable dimensions in developing countries. Appendix C reports a similar exercise for U.S. counties: within counties, manufacturing and service households have similar fertility and child enrollment rates, while counties with larger manufacturing exposure differ substantially in both outcomes.

to have been primarily consumer-facing throughout the transition, making it better suited for a calibration that covers many decades.

Second, the strict classification allows me to measure the mechanism directly in worker-level Census data. The model's key production-side object is not simply whether a sector purchases more finance or professional services. It is whether the producer-service tasks associated with that sector are high-skill intensive. With a strict industry classification, I can observe this object directly: within each broad industry group, I restrict attention to producer-service occupations and compute the share of workers with at least one year of college. If instead finance or professional services were split into a consumer-facing and a firm-facing component using expenditure weights, the corresponding worker-level skill moments would no longer be directly observable. The workers in finance serving final consumers may differ systematically from the workers in finance serving manufacturing firms. The baseline Census-based strategy is designed precisely to capture those differences in worker skill content rather than proxy for them with industry-level expenditure shares. Appendix E explains these classifications in detail and performs robustness exercises changing the definitions of consumer and producer services, the results remain qualitatively stable across multiple definitions of the consumer services bundle and calibration strategies.

The targeted moments fall into four groups. First, I target the time-series paths of agricultural, manufacturing, and service employment shares, fertility, and normalized GDP per capita. These moments discipline the reallocation of labor across sectors, the demographic transition, and the aggregate income path. Second, I target three scalar moments that summarize the transition: the peak manufacturing employment share, average fertility, and the agricultural-to-service fertility ratio. Third, I target final-period skill moments in 2019: the aggregate skilled employment share and the skilled employment shares within value added in agriculture, manufacturing, and consumer services. These moments discipline the supply of skilled labor and the sectoral skill intensities in value added production. Fourth, I target the producer-service moments that are most directly tied to the mechanism: the high-skill employment share within producer-service inputs used by agriculture, manufacturing, and consumer services in 2019, and the change in the aggregate high-skill producer-service share between 1960 and 2019.

The producer-service targets are central. I construct them from the 2019 IPUMS USA Census sample by first classifying workers into broad industries—agriculture, manufacturing, and consumer services—and then restricting attention within each broad industry to producer-service occupations. I then compute the share of those producer-service workers with at least one year of college. The empirical object is therefore not whether all manufacturing workers are more educated than all service workers. Rather, it is whether the producer-service occupations used by manufacturing are more education-intensive. This is the object mapped into the model's high-skill share within producer-service inputs. Appendix E describes the industry and occupation

classification, the IPUMS construction, and an international validation exercise showing that the same pattern is not unique to the United States.

Appendix E.7 considers an alternative input-output-based calibration of the same mechanism. In that exercise, I replace the baseline Census-based producer-service moments with moments constructed from OECD input-output tables. Specifically, I compute the high-skill service share induced by a \$100 final-demand shock to agriculture, manufacturing, and consumer services, and recalibrate the codification parameters to match those induced service mixes. I consider both the strict consumer-service definition used in the baseline and a final-expenditure-weighted definition in which mixed-use services such as finance can partly enter the consumer-service bundle. This robustness exercise is useful because it asks whether the results survive a different empirical discipline on the codification parameters. It is, however, less direct than the baseline calibration because it proxies high-skill services using service subsectors rather than directly observing the education of workers performing producer-service occupations.

The calibration proceeds in two steps. I first fix a set of parameters externally, either by normalization or by reference to the literature. I then choose the remaining parameters jointly by simulated method of moments. The model is simulated at an annual frequency and aligned so that model period $t_0 = 60$ corresponds to 1850. Appendix E reports the full target vector, data sources, numerical procedure, and robustness exercises.

5.1 Externally set parameters

Table 5 reports the parameters fixed outside the simulated method of moments. Manufacturing is the numeraire, unskilled efficiency is normalized to one, and initial productivity levels are normalized to zero in logs. I set a model generation to 20 years, following the quantitative unified-growth literature and the parameterization in [Lagerlöf \(2006\)](#). Final demand follows the non-homothetic CES structure of [Comin et al. \(2021\)](#); I set the elasticity of substitution across final goods to $\sigma = 0.3$, reflecting low substitutability across broad consumption categories.

The skill-choice parameters are set to give an interpretable transition from a low-skill to a high-skill economy. I normalize $\beta_\pi = 1$, which fixes the scale of the skill-choice index. The logit dispersion parameter σ_{skill} governs how changes in the log skill premium translate into changes in the probability of becoming skilled. Since

$$\log \left(\frac{p_{s,t}}{1 - p_{s,t}} \right) = \frac{\log \pi_t - c_0}{\sigma_{\text{skill}}},$$

the dispersion can be written as

$$\sigma_{\text{skill}} = \frac{\Delta \log \pi_t}{\Delta \log (p_{s,t} / (1 - p_{s,t}))}.$$

Table 5: Externally Set Parameters and Normalizations

Parameter	Value	Source / rationale
$p_{m,t}$	1	Manufacturing numeraire
h_u	1	Normalization
h_s	1.6	Fixed skill efficiency premium
$\log A_{j,0}$	0	Initial productivity normalization
gen	20	Generation length, Lagerlöf (2006)
ε_m	1	Manufacturing income-effect normalization
w_m	3.5	Manufacturing demand-weight normalization
σ	0.3	Final-goods elasticity, Comin et al. (2021)
β_π	1	Skill-choice index normalization
σ_{skill}	0.1	Benchmark logit dispersion
σ_L	1.2	Near-unit skill substitution elasticity
σ_{ps}	1.2	Near-unit producer-service mode elasticity
τ_e^U	1	Education-time cost normalization
τ_e^S	0.5	Benchmark value
τ_m, τ_s	1, 1	Child-rearing cost normalization
t_0	60	Model period aligned with 1850
$\theta_{j,1850}$	1	Initial value-added production
t_{GO}	2025	IO coefficients reach modern values

Notes: Appendix E reports the remaining fixed numerical parameters.

The U.S. calibration moves from an initial skilled share of approximately 0.02 to a terminal skilled share of 0.34. This implies

$$\Delta \log \left(\frac{p_s}{1 - p_s} \right) = \log \left(\frac{0.34}{0.66} \right) - \log \left(\frac{0.02}{0.98} \right) \simeq 3.23.$$

A long-run increase in the log skill premium of about 0.30 log points therefore implies

$$\sigma_{\text{skill}} \simeq \frac{0.30}{3.23} = 0.093.$$

I use $\sigma_{\text{skill}} = 0.1$ as the benchmark value. Thus, the parameter is chosen to discipline the speed of the skill-acquisition transition implied by historically plausible changes in the skill premium, rather than taken as a directly estimated primitive.

On the production side, I set the elasticities of substitution across skill types and producer-service modes to 1.2. These near-unit elasticities allow imperfect but nonzero substitution across inputs and are consistent with quantitative work that treats skill groups and tasks as imperfect substitutes in skill-complementary production structures ([Krusell et al., 2000](#); [Acemoglu and Autor, 2011](#)).

The education-time cost parameters govern the productivity of education investment in the household problem. I normalize $\tau_e^U = 1$, which fixes the units of education time in an unskilled environment, and set $\tau_e^S = 0.5$. I interpret $\tau_e(S_t) = (1 - S_t)\tau_e^U + S_t\tau_e^S$ as a reduced-form education-environment complementarity, not as tuition, public education expenditure, or a price index for schooling. A lower $\tau_e(S_t)$ means that a unit

of parental education time produces more effective child human capital in environments with more educated parents, skilled peers, teachers, and schooling institutions. The human-capital formation literature motivates $\tau_e^S < \tau_e^U$: child skills are produced through a technology with self-productivity and dynamic complementarity, so existing skill stocks and parental environments raise the productivity of later investments (Cunha and Heckman, 2007; Cunha et al., 2010). Although $\tau_e^S = 0.5$ implies that education time would be twice as productive in a fully skilled environment, the effective change along the calibrated U.S. path is much smaller. With a terminal skilled share of $S_t = 0.34$, the implied education-time cost is $\tau_e(0.34) = 1 - 0.5 \times 0.34 = 0.83$, a 17 percent reduction relative to an unskilled environment. I therefore view the benchmark as a conservative calibration of the education-environment complementarity.

I also allow input-output coefficients to evolve over the transition. The economy starts in 1850 with value-added production, $\theta_{j,1850} = 1$, and input-output coefficients gradually converge to their modern values by 2025.⁹ This choice is motivated by evidence that changes in U.S. industry output reflect changes in input-output coefficients as well as changes in final demand (Feldman et al., 1987).

5.2 Internally calibrated parameters

The remaining parameters are chosen jointly. Let

$$\Theta = \left(\gamma, \tau_a, \varphi_a, \varphi_m, \varphi_c, g^{GO}, \kappa_a, \kappa_m, \kappa_c, \eta_H, c_{col}, \alpha_a, \alpha_m, \alpha_c, \varepsilon_a, \varepsilon_c, w_a, w_c \right)$$

denote the internally calibrated vector. These parameters govern fertility preferences, sectoral fertility differences, sectoral productivity growth, gross-output productivity growth, producer-service codification, the cost of becoming skilled, sectoral skill intensities, and non-homothetic demand. Table 6 reports the calibrated values. The calibration is joint, but the table associates each parameter with the moment that most directly disciplines it.

The different groups of moments identify different parts of the model. Employment shares and final-demand moments discipline the non-homothetic demand parameters and the size of the manufacturing hump. Fertility moments discipline the household block and sectoral differences in child-rearing costs. Final-period skilled employment shares discipline the cost of becoming skilled and the skill intensities of value added. Sectoral productivity growth rates discipline the scale effects in productivity. The producer-service moments discipline κ_j , the sector-specific codification resistance parameters. These parameters are the main calibrated objects for the mechanism.

The calibration implies

$$\kappa_m < \kappa_c < \kappa_a.$$

9. I construct the modern coefficients from BEA input-output tables by mapping detailed industries into agriculture, manufacturing, consumer services, and producer services, and by computing sector-specific shares of value added, agricultural intermediates, manufacturing intermediates, and producer-service intermediates in gross output. Appendix E describes the mapping and construction in detail.

Thus, manufacturing related producer-service tasks are the easiest to reorganize into high-skill form, followed by producer-service tasks used by consumer services and agriculture. This ordering is not imposed directly. It is disciplined by the 2019 high-skill employment shares within producer-service inputs by final sector. In the calibrated economy, manufacturing matters not only because it employs workers directly, but because its intermediate-service needs are especially connected to high-skill organizational and technical tasks.

Table 6: Internally Calibrated Parameters

Parameter	Description	Value	Main target	Data	Model
γ	Utility weight on children	0.044	Average fertility	2.474	2.395
τ_a	Agricultural child cost	0.419	Fertility ratio A/S	2.000	1.977
φ_a	Agriculture scale effect	0.931	Agriculture TFP growth	0.037	0.040
φ_m	Manufacturing scale effect	0.828	Manufacturing TFP growth	0.031	0.030
φ_c	Consumer-services scale effect	0.528	Consumer-services TFP growth	0.014	0.012
g^{GO}	Gross-output TFP growth	0.003	GDPpc path deviation	0.000	0.145
κ_a	Codification resistance, A	1.007	HS in PS A, 2019	0.39	0.42
κ_m	Codification resistance, M	0.415	HS in PS M, 2019	0.65	0.71
κ_c	Codification resistance, C	0.731	HS in PS C, 2019	0.53	0.57
η_H	Frontier curvature	4.876	Δ aggregate HS in PS	0.182	0.168
c_{col}	skill acquisition cost	0.271	Skill share, 2019	0.345	0.308
α_a	Skill intensity, A	0.180	Within-VA skill A	0.160	0.161
α_m	Skill intensity, M	0.254	Within-VA skill M	0.220	0.229
α_c	Skill intensity, consumer services	0.237	Within-VA skill C	0.200	0.213
ε_a	Agriculture income effect	0.016	Average agriculture share	0.259	0.256
ε_c	Consumer-services income effect	1.087	Average services share	0.560	0.537
w_a	Agriculture demand weight	0.952	Agriculture share, 1850	0.606	0.620
w_c	Consumer-services demand weight	4.035	Services share, 1850	0.231	0.204

Notes: A = agriculture, M = manufacturing, C = consumer services, PS = producer services, HS = high-skill. The calibration is joint; the target column reports the moment most closely associated with each parameter. Producer-service high-skill targets are constructed from the 2019 IPUMS USA Census sample by restricting to producer-service occupations within each broad industry and computing the share of workers with at least one year of college.

5.3 Estimation

Let m^{data} denote the vector of empirical moments and $m^{model}(\Theta)$ the corresponding simulated moments. I choose Θ to minimize

$$\mathcal{L}(\Theta) = \frac{1}{K} \sum_{k=1}^K \left(\frac{m_k^{model}(\Theta) - m_k^{data}}{\max\{|m_k^{data}|, \underline{s}_k\}} \right)^2 + \mathcal{P}(\Theta), \quad (34)$$

where $\underline{s}_k$ is a moment-specific scale floor. For strictly positive level variables, such as normalized GDP per capita and the agricultural-to-service fertility ratio, I use log residuals. The term $\mathcal{P}(\Theta)$ penalizes high-frequency oscillations in the skill premium and the aggregate skilled share. These penalties are not additional economic targets; they rule out parameter vectors that match isolated moments through implausibly volatile transition paths.

The minimization uses a global dual-annealing routine followed by a Powell local search. The model is overidentified: the number of targeted moments exceeds the number of internally calibrated parameters. Appendix E provides the full numerical

details. In particular, the half-saturation parameter in the codification frontier, $H_{1/2}$, is normalized relative to the simulated scale of knowledge accumulation. For each candidate parameter vector, I first run a probe simulation with a fixed value of $H_{1/2}$. I then set $H_{1/2}$ equal to the average of the final observations of the resulting H_t path, subject to lower and upper bounds. This prevents the arbitrary units of H_t from mechanically determining the location of the codification frontier.

5.4 Model fit

Figure 8 compares the calibrated model to the main U.S. transition moments. The model reproduces the broad path of structural change: agricultural employment declines, manufacturing rises and then falls, and services become the dominant sector. It also captures the long-run demographic transition, with fertility declining from high nineteenth-century levels to modern low fertility. These moments discipline the two aggregate transitions that the model is built to connect: the reallocation of labor across sectors and the movement from high fertility to higher human capital investment.

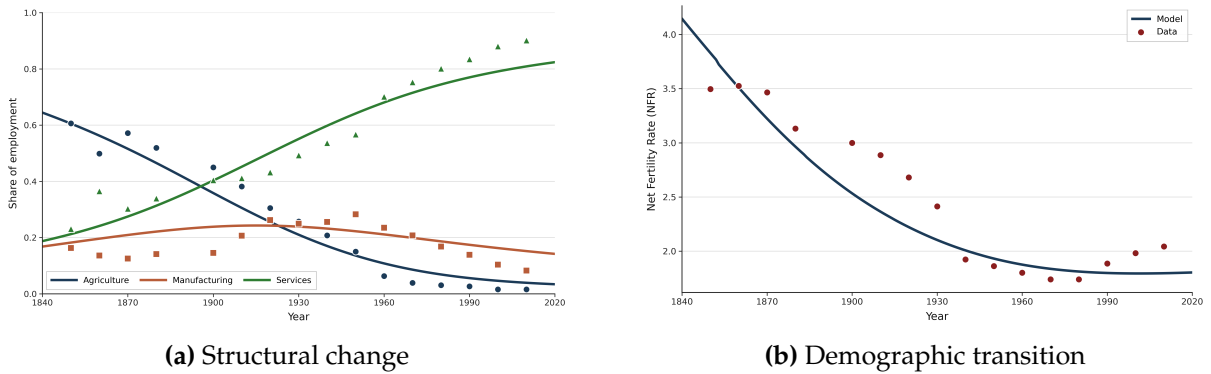


Figure 8: Model fit to targeted U.S. moments, 1850–2019. Dots represent data and lines show model simulations. Panel (a) shows employment shares in agriculture, manufacturing, and services. Panel (b) shows the fertility transition.

Figure 9 reports the fit for the moments most directly tied to the production side of the model. Panel (a) compares sectoral value added productivity growth rates. The model matches the ranking of growth rates across sectors, with faster productivity growth in agriculture and manufacturing than in services. Panel (b) compares the high-skill employment share within producer-service inputs by final sector. The model reproduces the central empirical ranking: producer services used by manufacturing are the most high-skill intensive, followed by those used by consumer services and agriculture. This ranking is the empirical counterpart of the model’s codification mechanism. Because the calibrated economy assigns the lowest codification resistance to manufacturing, manufacturing related producer-service tasks are the easiest to reorganize into high-skill form.

Table 6 summarizes the fit of the main scalar targets. The model matches average fertility closely, generating an average fertility rate of 2.395 compared with 2.474 in the data, and reproduces the agricultural-to-service fertility ratio, with a model value of

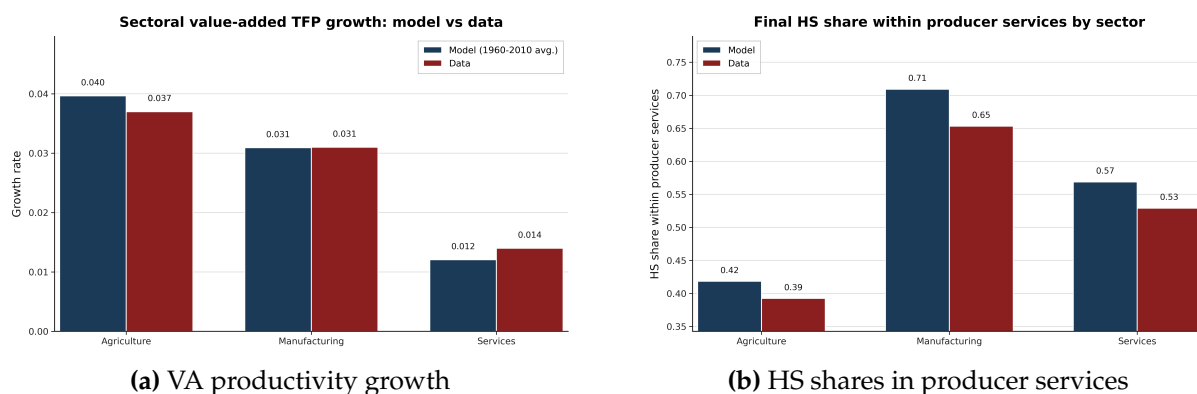


Figure 9: Model fit to mechanism-targeted moments. Panel (a) compares sectoral value added productivity growth rates in the model and the data. Panel (b) compares the final high-skill employment share within producer-service inputs by final sector. Data sources: GGDC 10-sector database for value added productivity growth and the 2019 IPUMS USA Census sample for high-skill employment shares within producer-service inputs.

1.977 compared with a target of 2.000. It also matches long-run sectoral productivity growth: agriculture, manufacturing, and services grow at 0.040, 0.030, and 0.012 in the model, compared with 0.037, 0.031, and 0.014 in the data. The model produces a final aggregate skilled share of 0.308, compared with 0.345 in the data. For the producer-service targets, the model matches the ranking across sectors and closely tracks the levels: 0.42 versus 0.39 in agriculture, 0.71 versus 0.65 in manufacturing, and 0.57 versus 0.53 in consumer services. The important discipline for the mechanism is that manufacturing related producer services are substantially more high-skill intensive than those used by the other final sectors, and the calibrated model reproduces this pattern.

Figure 10 reports the fit of normalized GDP per capita. GDP per capita is targeted over the calibration window, but matching this path is demanding because it is jointly shaped by sectoral productivity growth, labor reallocation, fertility, skill accumulation, and the evolution of input-output linkages. The calibrated model tracks the long-run increase in U.S. income and the acceleration of income growth over the development process.

I next examine non-targeted validation moments. Figure 11 compares model-implied GDP per capita growth and schooling to their empirical counterparts. These series are not directly targeted in the calibration. The model captures the broad decline in GDP per capita growth over development and generates a rising education path consistent with the long-run increase in schooling. In the model, schooling corresponds to the average education investment chosen by households; in the data, I compare it to average enrollment rates. This validation is informative because education responds endogenously to the growth-induced obsolescence mechanism rather than being targeted directly as a time series.

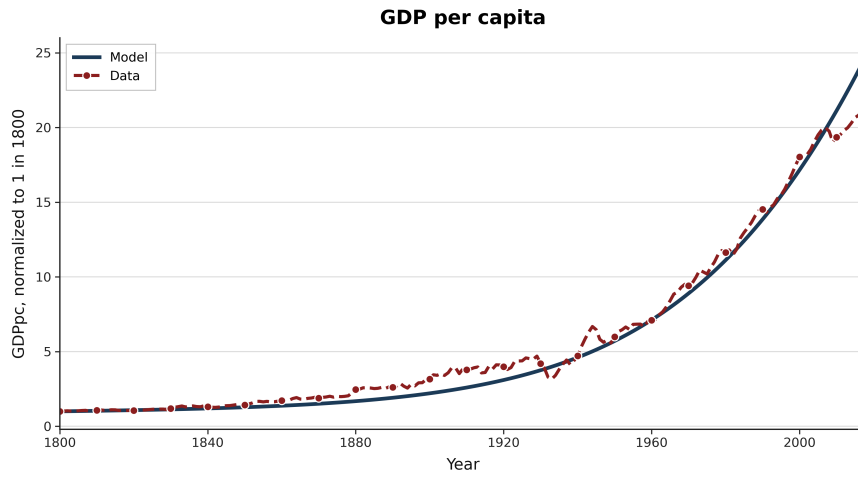
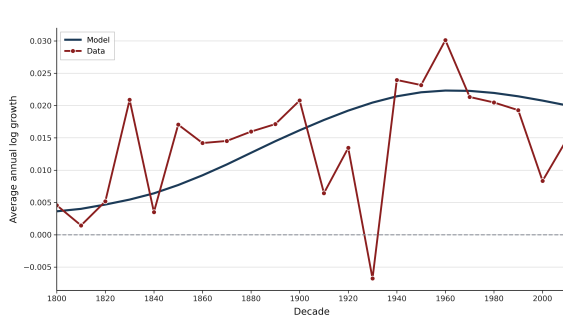
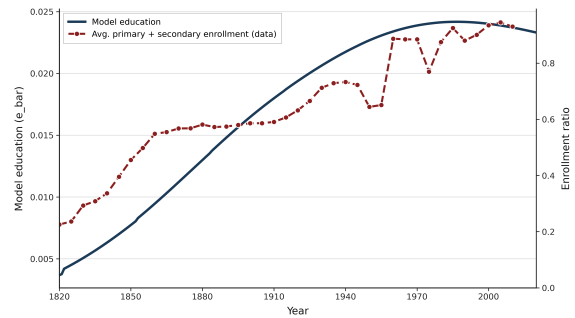


Figure 10: Model fit to normalized GDP per capita, 1850–2019. GDP per capita is normalized to one in 1850. Data source: Gapminder.



(a) GDP per capita growth



(b) Schooling

Figure 11: Non-targeted validation moments, 1800–2019. Panel (a) compares model-implied GDP per capita growth to the data. Panel (b) compares model education to enrollment rates. Schooling in the model is the average education choice across sectors; in the data it is the average enrollment rate, measured as the average of primary and secondary enrollment. Data sources: Gapminder and U.S. Censuses of Population.

As a final validation exercise, I compare the model and the data on the relationship between peak manufacturing employment and observable development outcomes. Figure 12 reports coefficients from regressions of key outcomes on the peak manufacturing employment share, estimated both in the model and in the data. These slopes are not used to estimate the U.S. calibration. The model nevertheless reproduces the qualitative pattern in the data: economies with larger manufacturing peaks have higher college attainment, larger high-skill service shares, faster GDP per capita growth, and faster demographic transitions. This exercise connects the calibrated U.S. mechanism back to the cross-country and county level evidence in Sections 2 and 3.

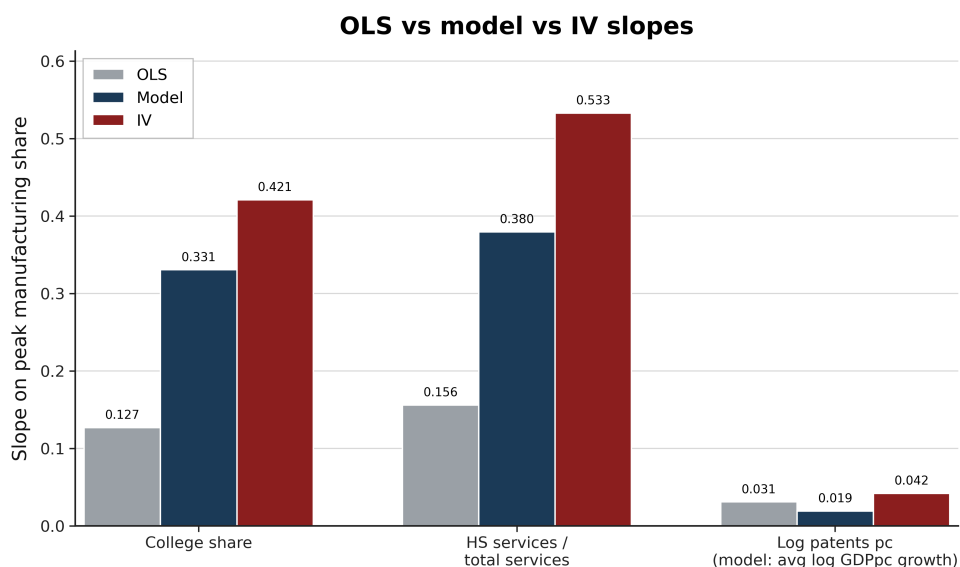


Figure 12: Non-targeted validation: model and data slopes with respect to peak manufacturing employment. The figure compares coefficients from regressions of key observable outcomes on the peak manufacturing employment share in the model and in the data.

The calibrated model therefore reproduces the main U.S. transition and the mechanism-specific moments needed for the counterfactual analysis. It matches the size and timing of the manufacturing phase, the demographic transition, the long-run ranking of sectoral productivity growth, and the high-skill intensity of manufacturing related producer services. It also generates non-targeted movements in schooling and GDP per capita growth, and reproduces the qualitative cross-economy relationship between peak manufacturing and development outcomes. I now use this calibrated economy to study how changing the size of the manufacturing phase affects long-run development.

6 Results

This section uses the calibrated U.S. economy to study how the size of the manufacturing phase affects long-run development. I generate counterfactual economies by varying the manufacturing demand weight w_m in the non-homothetic final-demand

system. A higher w_m raises the relative demand for manufacturing during the transition and produces a larger manufacturing employment peak. I then ask how this temporary difference in the path out of agriculture affects income, growth, education, population, and the composition of services.

The exercise is not meant to evaluate a specific policy instrument. It is a comparative-static exercise that varies the size of the manufacturing phase while holding fixed the rest of the calibrated economy. Appendix D shows that the same allocations can be implemented through relative-price wedges. Appendix F studies a more structural source of lower manufacturing exposure, through openness to trading partners with a comparative advantage in manufacturing. The goal here is simpler: isolate whether a smaller manufacturing phase leaves the economy with a different service sector and a different long-run development path.

Because the mechanism operates through producer services, the counterfactuals depend on the calibrated codification parameters $\kappa_a, \kappa_m, \kappa_c$. The baseline calibration disciplines these parameters using direct Census evidence on the education of workers in producer-service occupations within agriculture, manufacturing, and consumer services. Appendix E.7 shows that the results are robust to an alternative discipline based on OECD input-output spending shocks. In that exercise, I recalibrate only $\kappa_a, \kappa_m, \kappa_c$ to match the high-skill service content induced by final-demand shocks to agriculture, manufacturing, and consumer services, and then repeat the same manufacturing peak counterfactual. The qualitative results are unchanged under both a strict consumer-service definition and a final-expenditure-weighted definition. I therefore use the Census-based calibration as the baseline because it measures the model's task-level object more directly and is better suited to a long historical transition.

6.1 The manufacturing phase in the calibrated economy

Figure 13 reports the main comparative statics. Outcomes are normalized to the benchmark U.S. calibration. Economies with larger manufacturing peaks reach higher GDP per capita and higher long-run growth. They also experience lower cumulative population growth.

The mechanism is the one embedded in the model. A larger manufacturing phase raises demand for high-skill producer-service tasks and expands codified organizational knowledge. This raises the return to education, shifts household choices from child quantity toward child quality, and leaves the economy with a more skill intensive service sector after manufacturing employment declines. Manufacturing therefore matters not because it remains large in the long run, but because it changes the skill base and organizational frontier of the service economy that follows.

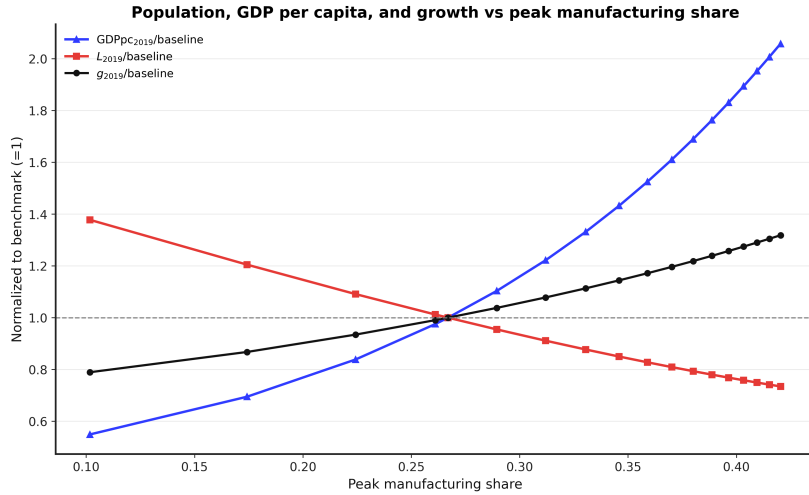


Figure 13: Comparative statics in the calibrated U.S. economy. The figure shows the relationship between the peak manufacturing employment share and GDP per capita, population, and long-run growth in 2019, with outcomes normalized to the U.S. benchmark. Counterfactual economies are generated by varying the manufacturing demand weight w_m in the final-demand system.

6.2 Accounting for cross-country patterns

I next ask whether the calibrated mechanism can account for the cross-country relationships documented in Section 2. For each country, I compute its observed peak manufacturing employment share and assign it the model economy with the corresponding peak, interpolating across the simulated grid. The exercise is deliberately one-dimensional: all countries are evaluated through the same calibrated economy, and the only source of variation is the size of the manufacturing phase. It is therefore not a full variance decomposition. Countries differ in institutions, geography, trade policy, natural resources, migration, education systems, demographic shocks, and measurement quality. The purpose here is narrower: to ask whether variation in the manufacturing phase alone generates the empirical gradients documented earlier in the paper.

Figure 14 reports the main comparison. The model generates the same qualitative pattern as the data across the four outcomes. Countries with larger manufacturing peaks are richer relative to the United States, grow faster over 2010–2019, have higher tertiary education, and experience lower cumulative population growth. The magnitudes are economically meaningful. For relative income, the data slope is 9.395 and the model slope is 2.946, so the manufacturing mechanism accounts for 31 percent of the cross-country relationship between peak manufacturing and log GDP per capita relative to the United States. For average GDP per capita growth, the model accounts for 60 percent of the corresponding slope, with a model slope of 2.201 compared with 3.669 in the data. This growth comparison should be interpreted as a long-run gradient rather than a forecast of growth in a particular decade, since realized growth over 2010–2019

also reflects business cycles, commodity prices, crises, and policy shocks outside the model.

The model also captures the human capital and demographic gradients emphasized throughout the paper. For tertiary education, the data slope is 0.762 and the model slope is 0.401, so the model accounts for 53 percent of the education gradient. For cumulative population growth, the data slope is -3.792 , while the model slope is -1.475 , so the model accounts for 39 percent of the cross-country relationship between peak manufacturing and population growth. This is the quantity-quality channel in the model: a larger manufacturing phase raises skill demand and growth, higher growth raises education investment, and higher education investment lowers fertility.

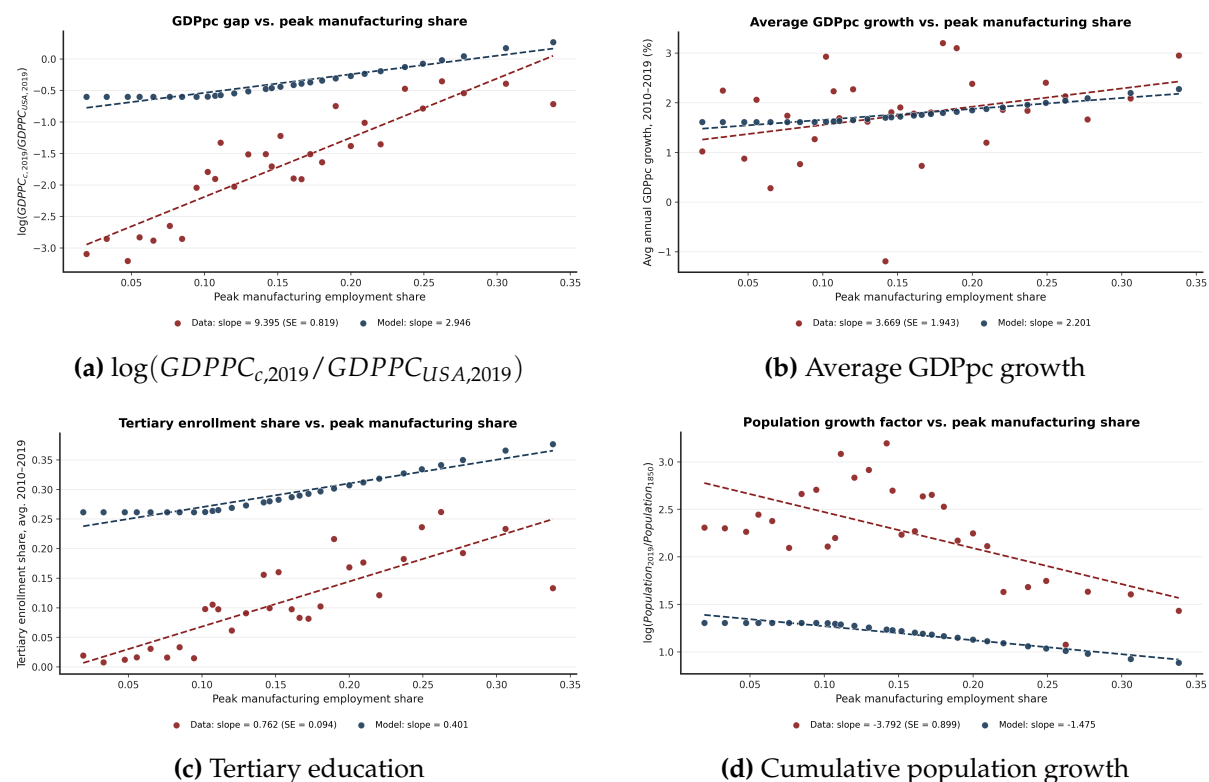


Figure 14: Peak manufacturing and development outcomes. The figure compares empirical cross-country relationships to model-generated counterfactual economies. Panel (a) plots final log GDP per capita relative to the United States. Panel (b) plots average GDP per capita growth over 2010–2019. Panel (c) plots tertiary enrollment over 2010–2019. Panel (d) plots $\log(Population_{2019}/Population_{1850})$. Red points and lines correspond to the data; blue points and lines correspond to model-generated economies.

The same mechanism also predicts that countries with larger manufacturing peaks develop more skill intensive service sectors. In the data, the FIRE share of service employment rises with peak manufacturing, with a slope of 0.385. In the model, the analogous object is high-skill producer-service employment as a share of service employment, and the corresponding slope is 0.677. The model therefore overshoots the empirical FIRE gradient. I interpret this comparison less literally than the income, education, and population results. In the model, high-skill producer-service work can be performed inside manufacturing firms, inside service firms, or in specialized

producer-service industries. In the data, I proxy for this object using employment in FIRE and professional-service industries. Since the boundary between in-house producer-service tasks and market producer-service industries varies across countries, the FIRE comparison is best read as evidence that the model generates the predicted shift toward high-skill producer-service activity, rather than as an exact accounting of a fixed industry category.

Table 7 reports country-level diagnostics that complement the slope comparisons. The slope ratios summarize how much of each empirical gradient the model generates. The correlations and associated R^2 values ask whether the model ranks countries correctly. The standard-deviation ratios ask whether the model generates as much cross-country dispersion as the data. These diagnostics give the same overall message. The model has meaningful predictive content for relative income, education, population growth, and the FIRE share, but the one-dimensional prediction is compressed relative to the full cross-country distribution for most outcomes. That compression is expected: the model varies one structural force, while the data combine many country-specific histories and shocks.

Table 7: Country-Level Model Fit

Outcome	N	σ_m/σ_d	Corr.	R^2	Slope ratio
GDP per capita	172	0.207	0.579	0.336	0.309
Tertiary enrollment	172	0.277	0.482	0.233	0.516
GDP per capita growth	172	0.081	0.127	0.016	0.606
Population growth	172	0.130	0.368	0.135	0.403
High-skill services	51	0.870	0.498	0.248	1.618

Notes: σ_m/σ_d is the ratio of model to data standard deviations. Corr. and R^2 compare country-level model predictions with the data. The slope ratio is the model slope divided by the data slope from separate regressions on peak manufacturing employment. All countries with non-missing data are included.

The quantitative message is not that manufacturing explains all cross-country dispersion. It does not, and the exercise is not designed to do so. The result is that the size of the manufacturing phase generates systematic and persistent differences in the service economy that follows. Countries with smaller manufacturing peaks can still move into services, but they do so with less human capital, weaker high-skill producer-service capacity, higher fertility, and lower long-run income. The quantitative message is not that manufacturing explains all cross-country dispersion. It does not, and the exercise is not designed to do so. The result is that the size of the manufacturing phase generates systematic and persistent differences in the service economy that follows. Countries with smaller manufacturing peaks can still move into services, but they do so with less human capital, weaker high-skill producer-service capacity, higher fertility, and lower long-run income.

6.3 Which mechanisms matter?

I next ask which calibrated sectoral heterogeneities account for the model's relationship between peak manufacturing employment and development outcomes. To do this, I run parameter-equalization experiments. In each experiment, I set one group of sectoral parameters equal to its arithmetic mean and repeat the manufacturing demand counterfactuals, leaving all endogenous choices and equilibrium responses active. Table 8 reports each experiment's slope as a percent of the full-model slope.

Table 8: Mechanism decomposition: slopes relative to the full model

	Full model	Equal κ_j	Equal φ_j	Equal α_j
GDPpc, 2019	100.0	64.9	77.8	97.7
GDPpc growth, 2010–2019	100.0	77.2	142.7	106.4
Tertiary education	100.0	43.3	65.6	105.5
Population growth	100.0	53.2	30.9	99.1
HS producer services	100.0	51.5	99.2	103.7

Notes: Entries report each experiment's slope as a percent of the full-model slope. Slopes are computed with respect to each experiment's own peak manufacturing employment share. In each equalization experiment, the relevant sectoral parameters are set equal to their arithmetic mean. κ_j governs codification resistance in producer-service tasks, φ_j governs sectoral productivity feedback, and α_j governs value added skill intensity. All endogenous model responses remain active. Outcomes are measured over the empirical transition window.

The decomposition points to producer-service codification as the key source of persistence. Equalizing κ_j reduces the 2019 income gradient to 65 percent of its full-model value and cuts the tertiary-education, population-growth, and high-skill producer-service gradients roughly in half. Thus, a large part of the manufacturing effect comes from the fact that manufacturing related producer-service tasks are easier to codify and reorganize into high-skill form. Equalizing φ_j also weakens the income, education, and population gradients, showing that sectoral productivity feedback shapes the transition through its effects on growth, education, and fertility. At the same time, it leaves the high-skill producer-service gradient essentially unchanged and *increases* the effect of manufacturing on the per capita growth rate in the final window. This is to be expected as the higher sectoral productivity growth of manufacturing, while increasing education which raises the codification frontier, also lowers population growth via the quantity-quality tradeoff which produces a drag on aggregate growth in the transition window.

By contrast, equalizing value added skill intensities, α_j , leaves the slopes close to their full-model values. The mechanism is therefore not mainly that manufacturing directly employs more skilled labor in value added. It operates through the organizational and producer-service margin: manufacturing expands the set of codifiable high-skill service tasks, raises the skill base accumulated during the transition, and changes the service economy that follows.

6.4 External validation: service linkages and subsequent growth

The main counterfactuals vary manufacturing demand inside the calibrated model. As an additional validation, I ask whether observed differences in service linkages across countries have predictive content outside the U.S. calibration. The exercise uses 1995 OECD input-output tables to compute, for each country, the high-skill service share induced by manufacturing demand and by consumer-service demand. I then match each country to the model economy with the closest service linkage structure and compute the model-implied GDP per capita gain from a standardized manufacturing expansion.

To connect the model exercise to realized development, I focus on countries that had not yet reached their manufacturing peak by 1995. For each country, I measure the subsequent expansion in manufacturing employment after 1995 and scale the model-implied gain by that realized expansion. The resulting object is a model-implied realized gain: how much GDP per capita the country should gain, according to the model, given its 1995 producer-service structure and its actual post-1995 manufacturing expansion. I then compare this model-implied gain to actual log GDP per capita growth over 1995–2019.

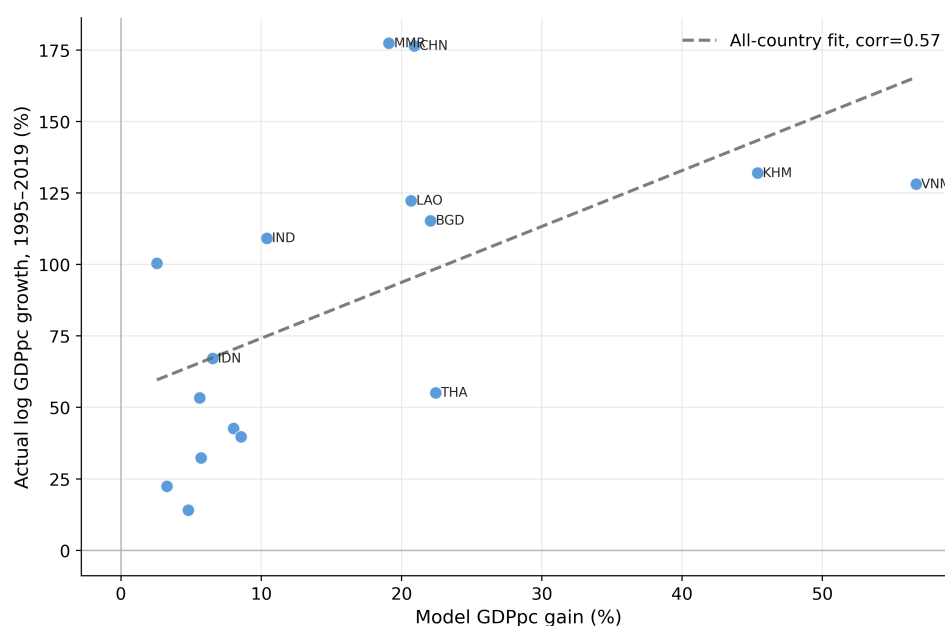


Figure 15: External validation: model-implied realized gains and subsequent GDP per capita growth. The sample includes countries that had not yet reached their manufacturing peak by 1995. The horizontal axis is the model-implied GDP per capita gain, obtained by matching each country’s 1995 input-output service linkage structure to the model and scaling the standardized model gain by the country’s realized post-1995 manufacturing expansion. The vertical axis is actual log GDP per capita growth over 1995–2019.

Figure 15 reports the result. Countries for which the model predicts larger gains from post-1995 manufacturing expansion tend to grow more afterward. This exercise is not a causal estimate: post-1995 growth reflects many forces besides manufacturing. Its

role is instead to validate the mechanism. Countries differ not only in how much manufacturing expands, but also in how strongly manufacturing is connected to high-skill producer services. The model predicts larger gains when those linkages are stronger, and the data show that this prediction has empirical content.

Taken together, the results support the central interpretation of the paper. Development without industrialization is possible in the sense that labor can leave agriculture and enter services. But the path out of agriculture changes the kind of service economy that emerges. A smaller manufacturing phase leaves less human capital, weaker high-skill producer-service capacity, slower fertility decline, and lower long-run income. Manufacturing is temporary in employment shares, but persistent in its effects on the skill base and organizational frontier of services.

7 Conclusion

This paper studies the long-run consequences of developing without industrialization. Labor can move out of agriculture and into services without a large manufacturing phase, but this transition need not produce the same service economy. Manufacturing creates demand for organizational, technical, managerial, and professional producer-service tasks that are intensive in skilled labor. These tasks help build the human capital base and organizational frontier of the service economy that follows.

The evidence points in this direction. Across countries, higher peak manufacturing employment shares are associated with faster growth after the transition into services, larger gains in tertiary education, faster fertility declines, and more skill intensive service sectors. Within the United States, counties with larger historical manufacturing booms later develop more education-intensive service occupations. I interpret these facts through a unified growth model with structural change, endogenous fertility, education, and heterogeneous producer services. In the model, manufacturing raises demand for producer-service tasks that are relatively easy to codify and reorganize into high-skill form. Calibrated to the United States over 1850–2019, the model accounts for 31 percent of the relationship between peak manufacturing and relative income, 60 percent of the relationship with GDP per capita growth, 53 percent of the relationship with tertiary education, and 39 percent of the relationship with cumulative population growth.

Development without industrialization is possible in the accounting sense that labor can leave agriculture and enter services. It is not neutral in the dynamic sense: bypassing manufacturing can leave the service economy with a lower organizational frontier. What matters is not only that services expand, but what kind of services expand. A path that bypasses manufacturing is more likely to end in a service economy concentrated in lower-skill consumer-facing activities. A path with a larger manufacturing phase is more likely to leave behind the skills, codified organizational knowledge, and high-skill producer-service capacity that support a higher-growth service economy.

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A Data Sources and Variable Construction

This appendix describes the data sources and variable construction used in the empirical analysis. The paper combines cross-country data on income, sectoral employment, fertility, and education with U.S. county level data on historical manufacturing activity, occupational skill content, geography, patents, education, income, and employment composition. Country-level variables are harmonized at the country-year level. Historical U.S. county level variables are harmonized to 1850 county boundaries unless otherwise stated.

A.1 Cross-country data

A.1.1 GDP per capita

Source note. Gapminder requests the citation: “GAPMINDER: GDP per capita v25 – constant PPP\$ 2011 based on World Bank, Maddison & IMF.” Documentation: <https://www.gapminder.org/sources/gdp-per-capita/>.

GDP per capita is from Gapminder’s GDP per capita series. The variable measures output per person in purchasing-power-parity adjusted international dollars. Gapminder constructs the series to make income levels comparable across countries and over time by adjusting for inflation and cross-country price differences. For recent decades, the series relies primarily on World Bank PPP GDP per capita data. For years before 1990, Gapminder uses historical income estimates from the Maddison Project and aligns these historical series to the modern PPP benchmark. For the most recent years in which World Bank observations are not yet available, Gapminder extends the series using IMF World Economic Outlook forecasts. The historical series is available back to 1800.

In the empirical analysis, I use the log of GDP per capita. Annual log GDP per capita growth is computed as

$$g_{c,t} = \log GDPpc_{c,t} - \log GDPpc_{c,t-1}.$$

For the main cross-country growth fact, the dependent variable is the average annual log GDP per capita growth rate over 2010–2019:

$$\bar{g}_{c,2010-2019} = \frac{1}{9} \sum_{t=2011}^{2019} (\log GDPpc_{c,t} - \log GDPpc_{c,t-1}).$$

I exclude 2020 from this exercise to avoid the effect of the COVID-19 shock.

A.1.2 Fertility

Source note. Fertility is from Gapminder’s “babies per woman” series, corresponding to the total fertility rate. Documentation: <https://www.gapminder.org/data/documentation/gd008/>.

Fertility is from Gapminder’s “babies per woman” series. The variable corresponds to the standard total fertility rate, defined as the average number of children a woman would have given current age-specific fertility rates. The series combines modern demographic data with historical estimates and extends back to 1800. I use the country-year total fertility rate to study the relationship between industrialization and the demographic transition.

For the cross-country fertility regression, the dependent variable is the average fertility rate over 1800–2020:

$$\overline{TFR}_c = \frac{1}{221} \sum_{t=1800}^{2020} TFR_{c,t}.$$

A.1.3 Sectoral employment shares and manufacturing peaks

Source note. Post-1990 sectoral employment shares are from Gapminder variables based on ILO modeled employment estimates. ILOSTAT documentation: <https://ilostat ilo.org/>. Historical manufacturing employment shares are supplemented using the GGDC 10-sector database: <https://www.rug.nl/ggdc/structuralchange/>.

Sectoral employment shares in agriculture, manufacturing, and services are taken from Gapminder/ILO for the post-1990 period and supplemented with the GGDC 10-sector database where available. The three broad sectors used in the paper are agriculture, manufacturing, and services.

The GGDC 10-sector database is used to extend manufacturing employment histories for countries covered by that database. In the GGDC data, the manufacturing employment share is constructed as

$$\text{Manu. Emp. Share}_{c,t} = \frac{EMP_{c,t}^{MAN}}{EMP_{c,t}^{SUM}},$$

where $EMP_{c,t}^{MAN}$ is manufacturing employment and $EMP_{c,t}^{SUM}$ is total employment in the GGDC 10-sector database. When both Gapminder/ILO and GGDC observations are available, I use the Gapminder/ILO series for the post-1990 harmonized employment shares and the GGDC series to fill earlier manufacturing observations.

The key cross-country treatment variable is the peak manufacturing employment share:

$$\text{Manu. Peak}_c = \max_t \text{Manu. Emp. Share}_{c,t}.$$

For countries with GGDC coverage, the peak is computed over the combined GGDC and Gapminder/ILO period. For countries without GGDC coverage, it is computed over the Gapminder/ILO period. The final regression sample contains 174 countries.

A.1.4 Education

Source note. Cross-country education is from the Barro-Lee educational attainment dataset. Data and documentation: <https://barrolee.com/>. The dataset is described in Barro and Lee (2013).

Cross-country education is from the Barro-Lee educational attainment dataset. I use the tertiary education measure matched to the country-year panel. Barro and Lee provide internationally comparable educational attainment measures for a large panel of countries at five-year intervals from 1950 to 2010 (Barro and Lee, 2013). In the cross-sectional figures, I average the tertiary education measure over 2010–2019 when observations are available.

For the long-difference education exercise, I compute the change in tertiary education between the first and last years in which both education and GDP per capita are observed:

$$\Delta e_c = e_{c,T_c} - e_{c,t_c},$$

and the corresponding change in log GDP per capita:

$$\Delta \log y_c = \log GDPpc_{c,T_c} - \log GDPpc_{c,t_c}.$$

Countries are then grouped by quintiles of $Manu. Peak_c$.

A.1.5 Cross-country controls

The main growth regressions include controls for average GDP per capita, the service employment share, years since the manufacturing peak, region fixed effects, and income-quintile fixed effects. Income quintiles are constructed from the cross-country distribution of GDP per capita. Years since the manufacturing peak are computed as the distance between the outcome period and the year in which country c reaches $Manu. Peak_c$.

A.2 Input-output data and service sector composition

Source note. The input-output exercise uses the 2015 OECD Inter-Country Input-Output table.

Data and documentation: <https://www.oecd.org/sti/ind/inter-country-input-output-tables.htm>.

The input-output exercise uses the 2015 OECD Inter-Country Input-Output table. I use the 2015 small ICIO table and aggregate country-industry cells to global industries. Let Z denote the matrix of intermediate flows and Y denote gross output. I construct the technical coefficient matrix

$$A_{ij} = \frac{Z_{ij}}{Y_j},$$

and compute Leontief total requirements using

$$x_j = (I - A)^{-1} f_j,$$

where f_j is a final-demand shock to sector j .

The exercise considers a \$100 final-demand shock to each of three broad using sectors: agriculture, manufacturing, and consumer services. Agriculture includes sectors A01, A02, and A03. Manufacturing includes all sectors whose OECD industry code begins

with C. Agriculture and manufacturing shocks are allocated across detailed industries in proportion to gross output.

The market-service universe and high-skill service classification are reported in Table 9. High-skill services are computer and information services, finance and insurance, and professional, scientific, and technical activities. All remaining market services are classified as other services.

Table 9: ICIO service sector classification

Category	OECD ICIO industries
Market services	<i>G, H49, H50, H51, H52, H53, I, J58T60, J61, J62_63, K, L, M, N, R, S</i>
High-skill services	<i>J62_63, K, M</i>

Notes: J62_63 denotes computer and information services, K denotes finance and insurance, and M denotes professional, scientific, and technical activities.

In the baseline exercise, consumer services are market-service industries for which final consumption exceeds 50 percent of gross output:

$$\frac{FinalConsumption_s}{GrossOutput_s} > 0.5.$$

Final consumption is measured using household final consumption expenditure, non-profit institutions serving households, and government final consumption columns in the ICIO table. The consumer-service shock is allocated across selected industries using observed final-consumption weights. In the robustness exercise, I instead allocate the consumer-services shock across all market services using observed final-consumption weights.

For each shock, I compute the induced service output and report the share of induced service output accounted for by high-skill services:

$$HSShare_j = \frac{\sum_{s \in HS} x_{j,s}}{\sum_{s \in Services} x_{j,s}}.$$

This is the object reported in the input-output figures. The purpose of the exercise is to compare the skill intensity of the service output induced by final demand in agriculture, manufacturing, and consumer services.

A.3 Historical U.S. county data

A.3.1 County harmonization

Source note. Historical county boundary shapefiles are from NHGIS/IPUMS. NHGIS documentation and data access: <https://www.nhgis.org/>.

All historical county level variables are harmonized to 1850 county boundaries. I construct geographic crosswalks using county shapefiles for each census decade and the 1850 county shapefile. For each source-year county, I compute the area overlap with 1850 counties using an equal-area projection and assign the source county to the 1850 county with the largest overlap. The resulting crosswalk maps decade-specific county identifiers to 1850 county identifiers. For variables observed on 2000 county boundaries, I allocate counts to 1850 counties using area weighted overlays.

A.3.2 Manufacturing peaks

Source note. County manufacturing data come from the Haines historical county dataset distributed by ICPSR, study 02896: <https://www.icpsr.umich.edu/web/ICPSR/studies/02896/staff>.

County manufacturing data come from historical Census of Manufactures county tabulations compiled in the Haines county dataset. I use the 1880–1940 censuses to construct the manufacturing peak because these are the most reliable county level manufacturing tabulations. The 1910 Census of Manufactures did not provide a complete county tabulation; it tabulated cities instead, so 1910 is omitted from the county manufacturing panel.

The county manufacturing measure is the number of manufacturing workers divided by total population:

$$\text{Manu. Emp. Share}_{c,t} = \frac{\text{Manu. Workers}_{c,t}}{\text{Population}_{c,t}}.$$

The county manufacturing peak is

$$\text{Manu. Peak}_c = \max_{t \in \{1880, 1890, 1900, 1920, 1930, 1940\}} \text{Manu. Emp. Share}_{c,t}.$$

If a county reaches the maximum in multiple years, I assign the first peak year:

$$T_c^{\text{peak}} = \min \left\{ t : \text{Manu. Emp. Share}_{c,t} = \text{Manu. Peak}_c \right\}.$$

A.3.3 Historical census sample and occupational skill content

Source note. Individual-level historical census data are from IPUMS USA full count U.S. Census files. Data and documentation: <https://usa.ipums.org/usa/>. Occupational education scores use the 1950 IPUMS occupation-based education score variable.

The historical event-study analysis uses full count U.S. Census data from 1850 to 1940. The baseline sample consists of fathers of children between ages 6 and 18 with non-missing county, industry, and employment information. This sample is used to construct county-year measures of school enrollment, fertility, sectoral employment, and occupational skill content.

I map industries into broad sectors using the IPUMS 1950 industry classification. Manufacturing is defined as industries with codes between 306 and 499. Services are defined as industries with codes between 506 and 946. Agriculture is the omitted primary-sector category.

Individual education is not observed in the pre-1950 full count censuses. I therefore measure the skill content of occupations using the IPUMS occupation education score from the 1950 Census. For each occupation, this score measures the share of workers in that occupation in 1950 who had completed at least some college. I assign this fixed 1950 occupation score to workers in earlier census years and average it within county-year-sector cells. The main outcome in the county event study is the mean education score of service sector occupations:

$$EDSCOR50_{c,t}^{serv} = \frac{1}{N_{c,t}^{serv}} \sum_{i \in (c,t,serv)} EDSCOR50_{occ(i)}.$$

This variable measures whether a county's service sector is composed of occupations that later appear as high-education occupations in the 1950 Census.

Treatment groups are based on the cross-county distribution of $Manu_Peak_c$. Event time is measured relative to T_c^{peak} . The event-study specification is estimated using the Borusyak, Jaravel, and Spiess imputation estimator (Borusyak et al., 2024). The implementation uses county and year fixed effects and the default standard-error procedure in `did_imputation`.

A.4 Geographic characteristics

Source note. Coal-field locations are from the NHGIS/USGS coal-field shapefile. Physiographic provinces are from the USGS physiographic provinces shapefile. River and lake geographic files are from the GIS files used in the county level construction.

The geographic characteristics used in the IV-style exercises are measured from 1850 county centroids. Distances are measured in kilometers and transformed as $\log(1 + d)$. Distance to coal is the distance from the 1850 county centroid to the nearest coal-field point. Coal-field locations come from the USGS/NHGIS coal-field shapefile, restricted to points within the 1850 county coverage. Distance to the fall line is computed from the boundary between the Piedmont and Atlantic Plain physiographic regions in the USGS physiographic provinces shapefile. I also use distance to the Ohio River and an indicator for whether the county is adjacent to the Great Lakes. These variables are merged into the county panel and interacted with the manufacturing peak decade in the instrument construction described in the main text.

A.5 Long-run county outcomes from NHGIS

Source note. Long-run county outcomes are constructed from 2000 Census county files downloaded from NHGIS. The relevant local extract files are `population_nhgis0030_ts_nominal_county.csv`, `nhgis0031_ds151_2000_county.csv`, `nhgis0032_ds153_2000_county.csv`, `nhgis0036_ts_nominal_county.csv`, and `nhgis0041_ds142_2000_county.csv`. NHGIS documentation: <https://www.nhgis.org/>.

I construct long-run county outcomes using 2000 Census data from NHGIS and harmonize them to 1850 county boundaries using area weighted overlays. Count variables are allocated by the area share of each 2000 county that overlaps an 1850 county. Mean income is first converted to an income sum by multiplying mean income by population, then allocated by area share and divided by allocated population.

College attainment is the fraction of the population age 25 and older with a bachelor's degree or graduate/professional degree. School enrollment is constructed from NHGIS counts of enrolled and not-enrolled school-age individuals. Per-capita income is constructed by allocating total income, defined as mean income times population, to 1850 county boundaries and dividing by allocated population.

Employment shares in agriculture, manufacturing, and services are constructed from 2000 NHGIS industry counts. I use the broad industry-by-sex counts from the 2000 Census. Agriculture combines employment in agriculture-related industries. Manufacturing combines manufacturing employment. Services combine transportation, information, finance, professional/business services, education/health, arts/accommodation, and other service categories.

For the long-run composition outcomes in Table 4, I also use the detailed 2000 NHGIS industry file `nhgis0041_ds142_2000_county.csv`. Table 10 reports the definitions of high-skill services, total services, innovation activities, advanced manufacturing, and legacy routine manufacturing.

Table 10: Long-run industry classifications

Variable	Included activities
High-skill services	Information; finance and insurance; real estate; professional, scientific, and technical services; management of companies and enterprises.
Total services	Information; finance and insurance; real estate; professional, scientific, and technical services; management; administrative support; education; health; arts; accommodation and food; other services.
Innovation activities	Data processing and information services; architectural and engineering services; computer systems design; management, scientific, and technical consulting; scientific research and development; computer and electronic products manufacturing; pharmaceutical manufacturing.
Advanced manufacturing	Computer and electronic products manufacturing; pharmaceutical manufacturing.
Legacy routine manufacturing	Textile mills; textile product mills; apparel; leather; wood; paper; printing; primary metals; fabricated metals; furniture manufacturing.

The high-skill services share is high-skill service employment divided by total service employment. The innovation employment share is innovation employment divided by total employment. The advanced manufacturing share is advanced manufacturing employment divided by total manufacturing employment.

A.6 Patents

Source note. Patent data are from PatentsView bulk data files. Data and documentation: <https://patentsview.org/download/data-download-tables>. The files used are `g_patent.tsv`, `g_ipc_at_issue.tsv`, `g_assignee_disambiguated.tsv`, and `g_location_disambiguated.tsv`.

Patent counts are constructed from PatentsView bulk data. I classify patents into broad technology sectors using IPC codes at issue. I construct the IPC subclass by combining IPC section, class, and subclass. Patents with an IPC subclass beginning with *A01* are classified as agriculture. Patents with IPC subclasses beginning with *G06*, *G07*, *G08*, or *H04* are classified as services, corresponding primarily to computing, data processing, control, and communications technologies. All remaining patents are classified as manufacturing. If a patent appears in multiple technology groups, I assign sectors using the priority order agriculture, services, and then manufacturing.

I then merge patents to assignee locations. PatentsView assignee records are linked to disambiguated location records using `location_id`. I keep assignee locations with non-missing geographic information and merge them to the patent table using `patent_id`.

Each patent-assignee location contributes one patent count. I then aggregate patent counts by county FIPS, state FIPS, and broad technology sector.

To merge patents to the historical county panel, I use the longitude and latitude attached to the PatentsView assignee location records. Patent-location observations are converted to point geometries and spatially joined to 1850 county polygons. Patent counts are then summed within 1850 counties:

$$\text{Patents}_c^k = \sum_{p \in c} \mathbb{1}\{\text{sector}(p) = k\}, \quad k \in \{A, M, S\}.$$

The main long-run outcome in Table 4 is log patents per capita, constructed using total patent counts and the 2000 population harmonized to 1850 county boundaries.

A.7 University openings

Source note. University and college openings are collected from Wikidata and supplemented with information from English Wikipedia pages. Wikidata query service: <https://query.wikidata.org/>.

I collect university and college openings from Wikidata. The query retrieves institutions with a founding year, coordinates, and an English Wikipedia page. I keep institutions with non-missing founding years and locations. Institutions are assigned to counties using their coordinates. For descriptive exercises, I group counties by their historical manufacturing peak and compute the distribution of university openings across peak-manufacturing groups.

I also classify institutions by topic using the first 250 words of their Wikipedia page. Topic categories include clergy, agriculture, liberal arts, commerce, business, law, mechanical arts, medicine, military, teacher training, and other.

B Additional Cross-Country Evidence

This appendix reports additional cross-country evidence supporting the facts documented in Section 2. The exercises below are not used in the calibration. They show that the relationship between peak manufacturing employment and development outcomes is robust to alternative specifications and that the input-output mechanism is not driven by the baseline definition of consumer services. Variable definitions and data sources are described in Appendix A.

B.1 GDP per capita growth without service employment controls

The baseline growth regressions in Section 2 control directly for the average service employment share over 2010–2019. This is a demanding specification because the service share is itself an outcome of structural change and is mechanically related to the

decline of agriculture and manufacturing. Table 11 reports the same set of regressions after omitting the service employment-share control.

Table 11: Dep. Var. is Avg GDPPC growth rate 2010-2020

	(1)	(2)	(3)	(4)
Peak Emp. % Man.	3.334* (1.83)	5.164** (2.10)	4.201 (1.54)	5.002* (1.71)
N	174	174	174	174
R2	0.0143	0.0495	0.105	0.130
Income Quintile FE	no	yes	yes	yes
Region FE	no	no	yes	yes
Controls	no	no	no	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The coefficient on Manu. Peak_c remains positive and is only marginally insignificant at the 10% level when controlling only for income quintile and regional fixed effects. Thus, the positive association between peak manufacturing employment and subsequent GDP per capita growth is not driven by directly conditioning on the size of the service sector. Instead, countries with larger manufacturing peaks grow faster even when the specification compares them without holding fixed the contemporaneous service employment share; that is, countries with a higher peak tend to grow faster even when compared to countries which are at an earlier stage of the development process which, *ceteris paribus*, would enjoy the advantages of catch up growth and grow faster than countries at the frontier.

B.2 human capital quality

Section 2 shows that countries with larger manufacturing peaks accumulate more tertiary education over development. This subsection reports a complementary exercise using human capital quality, measured by test scores. The dependent variable is human capital quality in 2015. The regressions control for average log GDP per capita over 2010–2019 and progressively add income-quintile fixed effects, region fixed effects, and additional controls.

Across specifications, Manu. Peak_c is positively associated with human capital quality. The relationship remains statistically significant after controlling for income levels, income-quintile fixed effects, regional fixed effects, and additional controls. This suggests that the link between manufacturing and human capital is not only about years of schooling or tertiary enrollment. Countries with larger manufacturing peaks also tend to have higher measured education quality.

Table 12: Dep. Var. is Human Capital Quality, 2015

	(1)	(2)	(3)	(4)
Avg. log GDP per capita, 2010–2019	0.618*** (5.36)	0.523*** (2.74)	0.571* (1.82)	0.462 (1.42)
Peak Emp. % Man.	4.943*** (3.97)	4.866*** (4.05)	3.554*** (2.91)	3.237*** (2.66)
N	79	79	79	79
R2	0.552	0.559	0.697	0.702
Income Quintile FE	no	no	yes	yes
Region FE	no	no	yes	yes
Controls	no	no	no	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

B.3 Input-output robustness: alternative consumer-service demand

The baseline input-output exercise in Section 2 defines consumer services as market-service industries for which final consumption exceeds 50 percent of gross output. This subsection reports a robustness exercise using a broader definition. Instead of selecting only consumer-oriented service industries, I allocate the consumer-services final-demand shock across all market services using observed final-consumption weights. This allows final consumers to purchase services that are also heavily used as intermediate inputs, such as finance, computer and information services, and professional services.

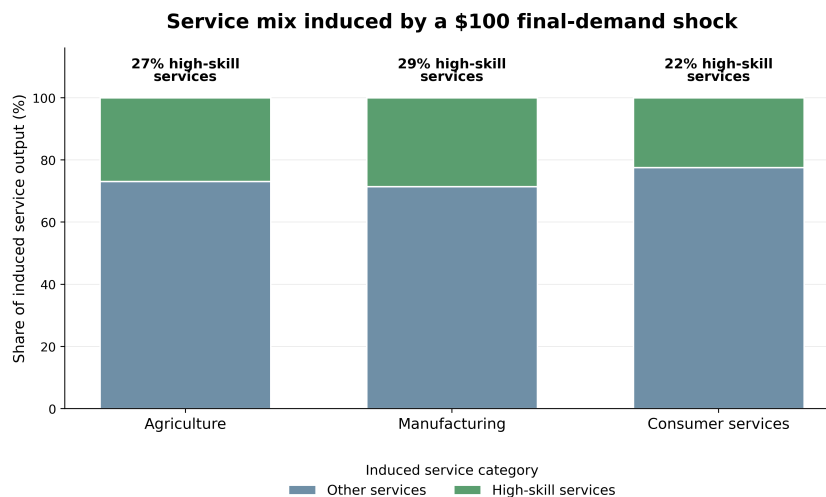


Figure 16: Robustness of the input-output service mix exercise. Bars show the service sector mix induced by a \$100 final-demand increase in agriculture, manufacturing, or consumer services, computed using Leontief total requirements from the 2015 OECD ICIO table. In this robustness exercise, the consumer-services shock is allocated across all market services using observed final-consumption weights.

The qualitative pattern is unchanged. Manufacturing induces a more high-skill intensive service mix than consumer-service demand, even when consumer-service demand is allowed to include high-skill market services. This supports the interpretation that manufacturing matters not only through direct manufacturing employment, but also through its intermediate-input linkages to high-skill producer services.

B.4 Input-output robustness by country income group

Figure 17 repeats the input-output service mix exercise by country income group. This checks whether the baseline pattern is driven by a small set of high-income economies or whether manufacturing intensive demand is associated with a more high-skill service mix more broadly.

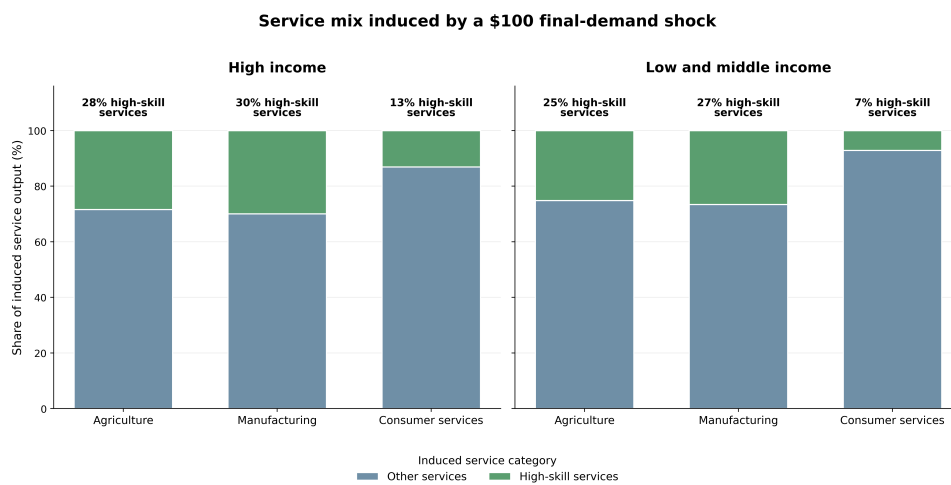


Figure 17: Input-output service mix exercise by income group. Bars show the service sector mix induced by a \$100 final-demand increase in agriculture, manufacturing, or consumer services, computed from Leontief total requirements using the 2015 OECD ICIO table. Consumer services are defined as service industries for which final consumption exceeds 50 percent of gross output.

The manufacturing shock remains relatively intensive in high-skill services across income groups. The magnitude varies across groups, but the broad ordering is consistent with the mechanism emphasized in the paper: manufacturing demand pulls on producer-service activities that are more skill intensive than the services generated by consumer-facing final demand.

C Additional U.S. County Evidence

This appendix provides additional evidence for the U.S. county analysis in Section 3. I first document the timing and distribution of historical manufacturing peaks across counties. I then report a robustness exercise that splits counties into finer bins of the manufacturing peak distribution. I next decompose the service sector education-score

result into the occupations that drive it. Finally, I provide complementary evidence from university openings and historical help-wanted advertisements.

C.1 Timing and size of county manufacturing peaks

Figure 18 summarizes the timing and size of county manufacturing peaks. For each county, I define Manu. Peak_c as the maximum manufacturing employment share observed in the Census of Manufactures county tabulations over 1880, 1890, 1900, 1920, 1930, and 1940. The 1910 Census of Manufactures is omitted because it does not provide complete county level manufacturing labor tabulations. The figure reports, by peak decade, the average peak manufacturing employment share among counties peaking in that decade and the number of counties reaching their peak in that decade.

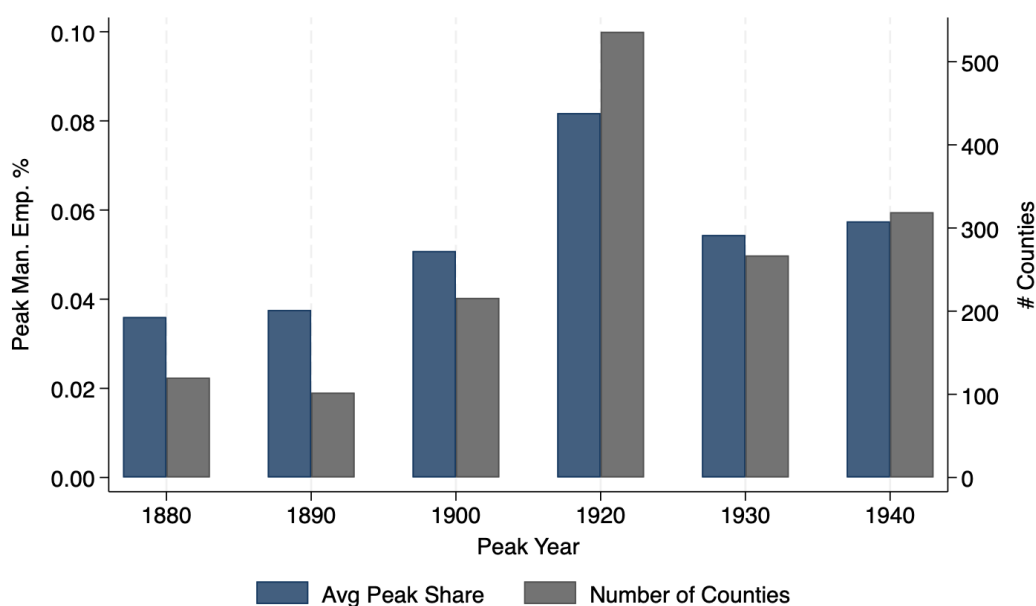


Figure 18: Timing and size of county manufacturing peaks. The figure reports the average value of Manu. Peak_c among counties peaking in each decade and the number of counties reaching their manufacturing peak in that decade. County manufacturing peaks are constructed from Census of Manufactures county tabulations over 1880, 1890, 1900, 1920, 1930, and 1940.

The distribution shows substantial variation in the timing of industrialization across counties. Some counties reach their manufacturing peak during the late nineteenth century, while others peak only in the interwar period. This variation is useful for the event-study design because treatment timing differs across counties while outcomes are observed in common census decades.

C.2 Panel evidence during the U.S. transition

Before turning to the event-study design, I examine whether manufacturing intensive counties experience different changes in fertility, schooling, and the skill content of

services during the U.S. transition from a high fertility/agrarian/low schooling economy to the highly educated and industrialized country it was by the mid 20th century. This exercise uses the full count Census sample from 1850 to 1940 and estimates panel regressions of the form

$$\Delta y_{ct} = \alpha + \beta \text{Manu. Emp. Share}_{c,t-1} + \Gamma' X_{c,t-1} + \varepsilon_{ct},$$

where y_{ct} is fertility, school enrollment, or the mean education score of service occupations in county c and decade t . The dependent variables are measured in first differences, and the main regressor is the lagged county manufacturing employment share. The controls include the urbanization rate, log population, and the agricultural employment share. The coefficient β therefore asks whether, conditional on broad measures of county development, more manufacturing intensive counties subsequently experience larger or smaller changes in the outcome. In the more demanding specifications, I also include state-by-year fixed effects to absorb state-level shocks and policy changes, including changes in compulsory schooling laws.

Table 13: Panel Data (1850–1940)

	(1) Δ Fertility	(2) Δ Enrollment	(3) Δ Ed. Score	(4) Δ Fertility	(5) Δ Enrollment	(6) Δ Ed. Score
L.% Man Emp.	-0.282*** (0.067)	0.109*** (0.035)	6.267*** (0.504)	-0.312*** (0.064)	0.086*** (0.025)	2.181*** (0.567)
State $\times$ Year FE	No	No	No	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12758	12758	12748	12758	12758	12748
Within R2	0.022	0.060	0.075	0.522	0.546	0.568

Standard errors are clustered at the county level. Dependent variables are measured in first differences. Data source: US Census of Population 1850-1940.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13 reports the results. Counties with higher lagged manufacturing employment shares subsequently experience larger fertility declines, larger increases in school enrollment, and larger increases in the education score of service occupations. The signs are stable across specifications with and without state-by-year fixed effects. The magnitudes decline once state-by-year fixed effects are included, but the relationships remain statistically significant. This evidence is descriptive, but it shows that the three outcomes emphasized in the paper—fertility decline, schooling, and the skill content of services—move together with manufacturing intensity within U.S. counties during the transition.

C.3 Alternative peak-manufacturing bins

The baseline event study in Figure 5 compares mutually exclusive groups of counties in the tails of the manufacturing peak distribution. Figure 19 reports a finer grouping. Counties are divided into seven mutually exclusive bins: the bottom decile, the

10th–20th percentiles, the 20th–30th percentiles, the middle 30th–70th percentiles, the 70th–80th percentiles, the 80th–90th percentiles, and the top decile of the Manu. Peak_c distribution.

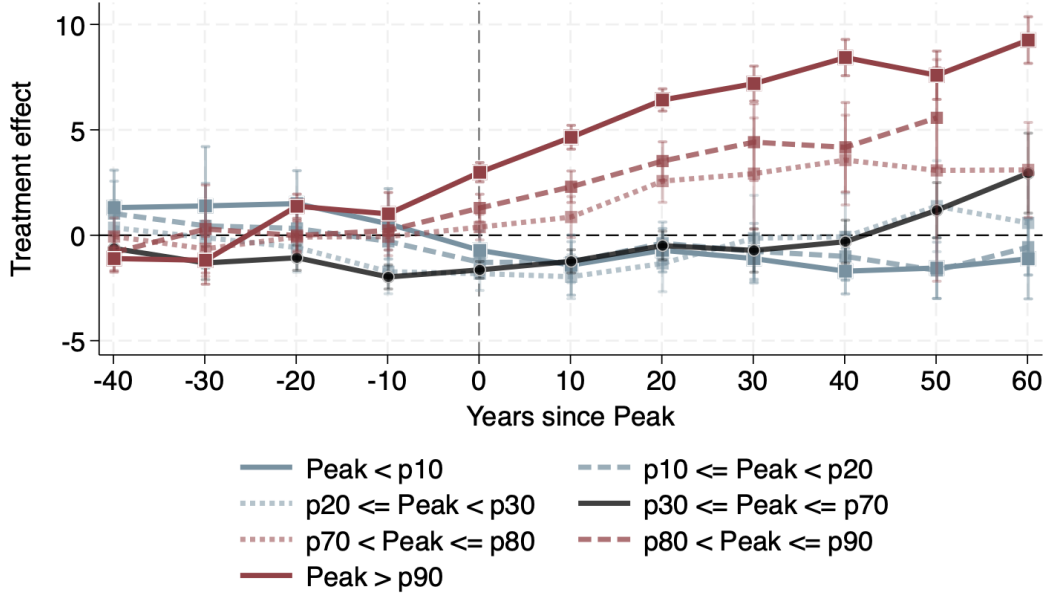
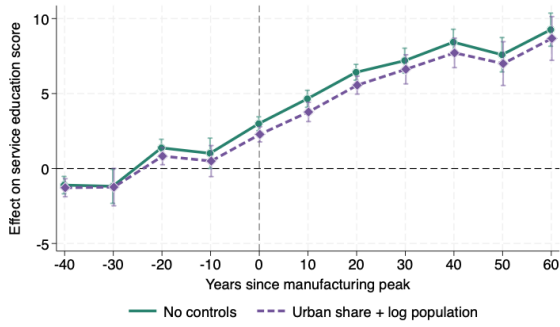


Figure 19: Event-study estimates by mutually exclusive peak-manufacturing bins. The outcome is the mean 1950 occupation education score among service sector workers in county c and decade t . Counties are grouped by their position in the cross-county distribution of Manu. Peak_c . Event time is measured in years relative to the county manufacturing peak. Estimates are obtained using the Borusyak, Jaravel, and Spiess imputation estimator with county and census-decade fixed effects.

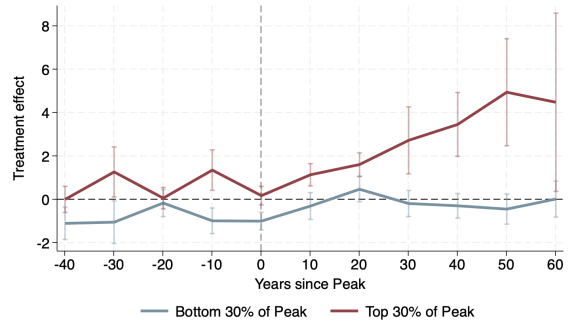
The results confirm that the baseline pattern is concentrated in the upper tail of the manufacturing peak distribution. Counties with the largest manufacturing peaks experience the strongest post-peak increase in the educational content of service occupations. Counties in the middle of the distribution display much smaller changes, while counties in the lower tail do not exhibit a comparable increase. This pattern supports the interpretation that the intensity of the manufacturing phase matters for the subsequent composition of the service economy.

C.4 Robustness to Urbanization, Market Size, and IV Specification

A natural concern is that the event-study estimates reflect the growth of larger and more urban counties rather than the legacy of manufacturing itself. manufacturing intensive counties may have become urban centers, and urban counties may have developed more educated service sectors for reasons unrelated to the organizational capabilities emphasized in the paper. A second concern is that the peak-manufacturing measure may be endogenous to local service sector trajectories. Figure 20 addresses these concerns by reporting two complementary robustness exercises.



(a) Urbanization and market-size controls



(b) IV event-study specification

Figure 20: Robustness of service sector event-study estimates. Panel (a) compares the top-decade manufacturing peak event study with and without controls for contemporaneous urban share and log county population. Panel (b) reports the IV event-study specification. The outcome is the mean 1950 occupation education score among service sector workers in county c and decade t . Estimates are obtained using the Borusyak, Jaravel, and Spiess imputation estimator with county and census-decade fixed effects.

Panel (a) compares the baseline event-study path for counties in the top decile of the manufacturing peak distribution with an otherwise identical specification that controls for contemporaneous urban share and log county population. Urban share is measured as the population living in places above 25,000 residents divided by total county population. Panel (b) reports the corresponding IV event-study specification. Across both exercises, the dynamic pattern remains similar to the baseline estimates, suggesting that the rise in service sector education scores is not explained by differential urbanization, county market size, or the direct endogeneity of the peak-manufacturing measure alone. Because urbanization and population growth may themselves be consequences of industrial development, the controls in Panel (a) are interpreted as a robustness check rather than as the preferred estimate of the total effect.

C.5 Which occupations drive the service sector education-score result?

The event-study outcome $EDSCOR50_{c,t}^{serv}$ increases when service employment shifts toward occupations that had higher college intensity in the 1950 Census. Figure 21 decomposes this change by occupation. I compare counties in the top and bottom deciles of the $Manu_{c,t}$ distribution and compute, for each occupation, the difference-in-differences in employment-share changes between 1850 and 1940. The dark blue bars report this occupation-level difference-in-differences in percentage points. The light blue bars report the occupation's 1950 education score.

The occupations with the largest positive differential growth in high-peak manufacturing counties include physicians and surgeons, clerical workers, stenographers and secretaries, lawyers and judges, teachers, accountants, office-machine operators, and

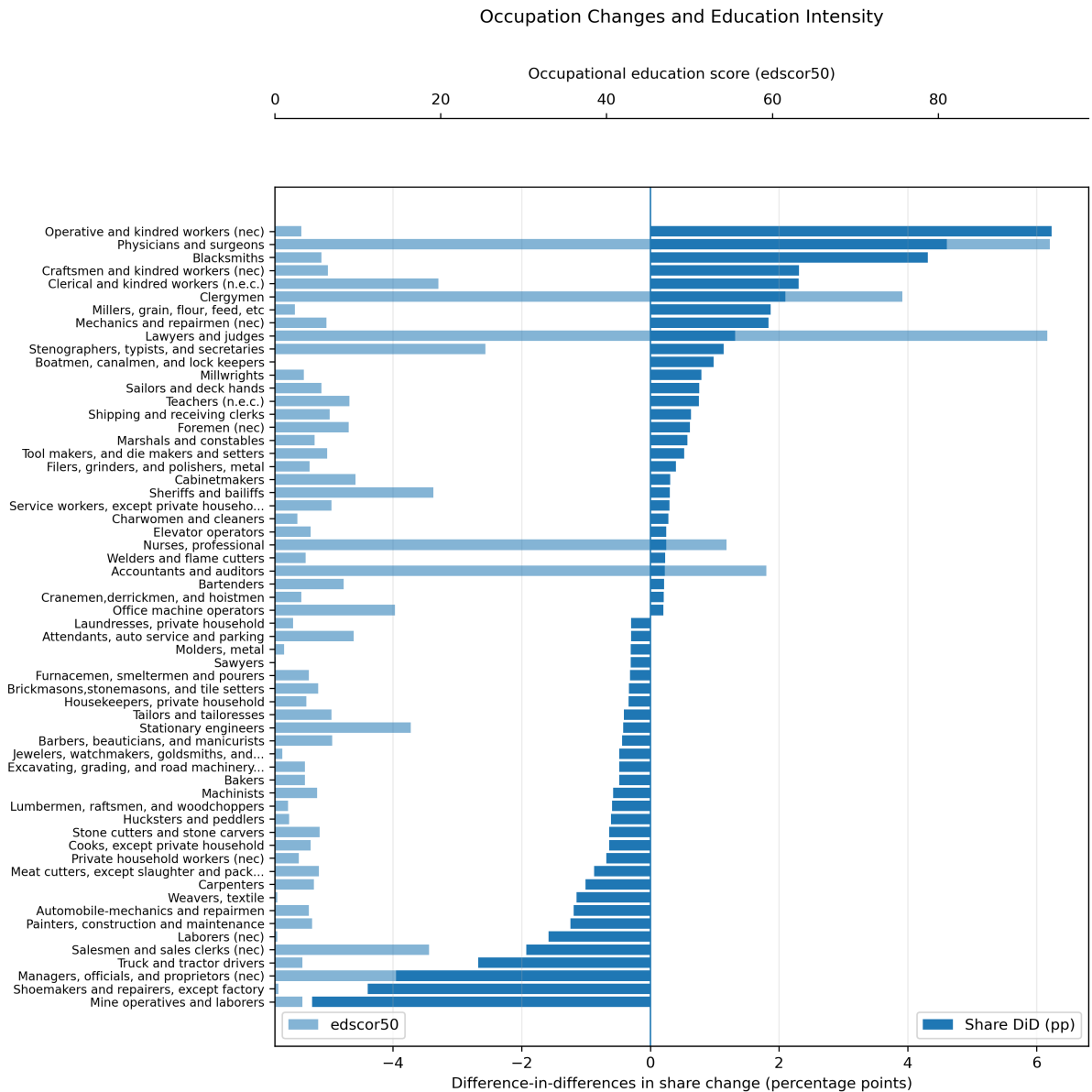


Figure 21: Occupation-level decomposition of changes in service sector education intensity. Blue bars report the difference-in-differences in occupation employment-share changes between high- and low-peak manufacturing counties from 1850 to 1950. Orange bars report the 1950 occupation education score, defined as the share of workers in the occupation with at least one year of college. Farm occupations are excluded.

other professional or white-collar occupations. These occupations are not manufacturing production jobs, but many are organizationally complementary to manufacturing expansion: they involve accounting, administration, legal services, communication, technical support, record keeping, and coordination. By contrast, occupations with negative differential growth include laborers, salesmen, shoemakers, operatives, and other lower-education manual or routine occupations. The decomposition therefore suggests that historical manufacturing peaks are associated with a later shift in services toward the kinds of professional and organizational occupations emphasized by the mechanism.

C.6 Service industries and occupations by education intensity

Tables 14 and 15 provide additional detail on the service industries and occupations with low and high educational content in the 1950 Census. These tables help interpret the education-score outcome by showing the types of activities that receive high and low *EDSCOR50* values.

Table 14: Service sub-industries by college intensity in 1950

Low college intensity		High college intensity	
Sub-industry	%	Sub-industry	%
Private households	3.5	Educational services	75.8
Shoe repair	4.3	Legal services	73.5
Trucking	5.3	Medical, excl. hospitals	71.2
Taxicabs	5.7	Welfare and religious services	59.1
Dressmaking	6.1	Engineering and architecture	58.4
Rail and bus transit	6.4	Misc. professional services	53.9
Misc. repair services	7.1	Drug stores	45.7
Sanitary services	7.2	Radio and television	43.0
Auto repair	8.0	Accounting and auditing	42.7
Water transport	8.1	Hospitals	38.5

Notes: Entries report the fraction of workers in each service sub-industry with at least one year of college in the 1950 Census.

Table 15: Service occupations by college intensity in 1950

Low college intensity		High college intensity	
Occupation	%	Occupation	%
Teamsters	0.0	Mathematicians	100.0
Boatmen, canalmen, lock keepers	0.0	Biological scientists	96.0
Bootblacks	0.0	Chiropractors	93.9
Telegraph messengers	0.0	Professors and instructors, n.e.c.	93.8
Midwives	0.0	Professors: engineering	93.8
Hucksters and peddlers	1.7	Professors: chemistry	93.8
Longshoremen and stevedores	2.1	Professors: biological sciences	93.8
Laundresses, private household	2.2	Professors: mathematics	93.8
Charwomen and cleaners	2.7	Professors: agricultural sciences	93.8
Private household workers, n.e.c.	2.9	Professors: physics	93.8

Notes: Entries report the fraction of workers in each occupation with at least one year of college in the 1950 Census. manufacturing heavy occupations are excluded from the low-college list.

The high-education service activities are concentrated in legal, medical, educational, accounting, engineering, architectural, professional, and technical occupations. This is consistent with the mechanism in the paper: manufacturing exposure predicts the later emergence of service activities that are complementary to organization, coordination, technical knowledge, and professional expertise.

C.7 University openings

I next examine whether counties with larger historical manufacturing peaks were also more likely to develop local higher-education institutions. University and college openings are collected from Wikidata, assigned to counties using institution coordinates, and grouped by founding year. Figure 22 plots the fraction of universities founded between 1860 and 2000 across counties grouped into quartiles of historical peak manufacturing intensity.

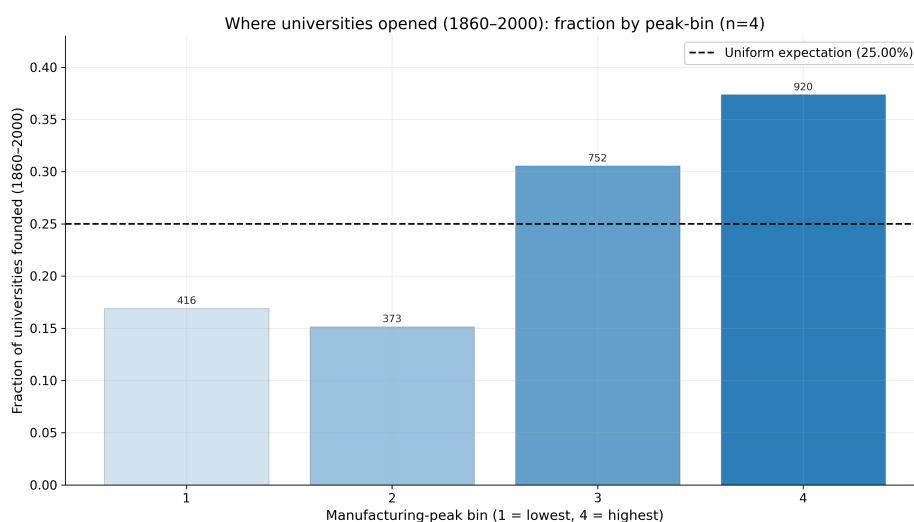


Figure 22: University openings and historical manufacturing exposure. The figure reports the fraction of universities founded between 1860 and 2000 across counties grouped into quartiles of Manu. Peak_c . University openings are collected from Wikidata and assigned to counties using institution coordinates.

Counties with larger manufacturing peaks account for a disproportionate share of university openings. This evidence is descriptive, but it is consistent with the idea that industrialization is associated with the local development of institutions that support skill formation and human capital accumulation.

C.8 Help-wanted advertisements and demand for skilled labor

I also use historical help-wanted advertisements to study whether manufacturing expansion is associated with demand for skilled labor. The data are constructed from the Library of Congress *Chronicling America* newspaper repository. The underlying corpus begins from pages containing a proximity match for “help wanted.” Newspapers are geolocated and grouped into geographic regions. Region-year outcomes are constructed from occupation keyword counts.

I distinguish between skilled manufacturing occupations and skilled service or white-collar occupations. Skilled manufacturing keywords include occupations such as machinist, millwright, and boilermaker. Skilled service and white-collar keywords include occupations such as clerk, bookkeeper, and stenographer. To normalize for

changes in newspaper coverage and general labor-market advertising intensity, I compare these keyword counts to a baseline basket of common lower-skill occupations: cook, waiter, janitor, and laborer.

I estimate an event study around the appearance of new or expanding factory mentions in local newspapers. A region is treated in the first year in which factory-expansion mentions exceed the threshold used in the text-classification procedure. Figure 23 reports the long-run event-study estimates. The dependent variable is the log of occupation keyword hits normalized by the baseline basket of lower-skill occupations.

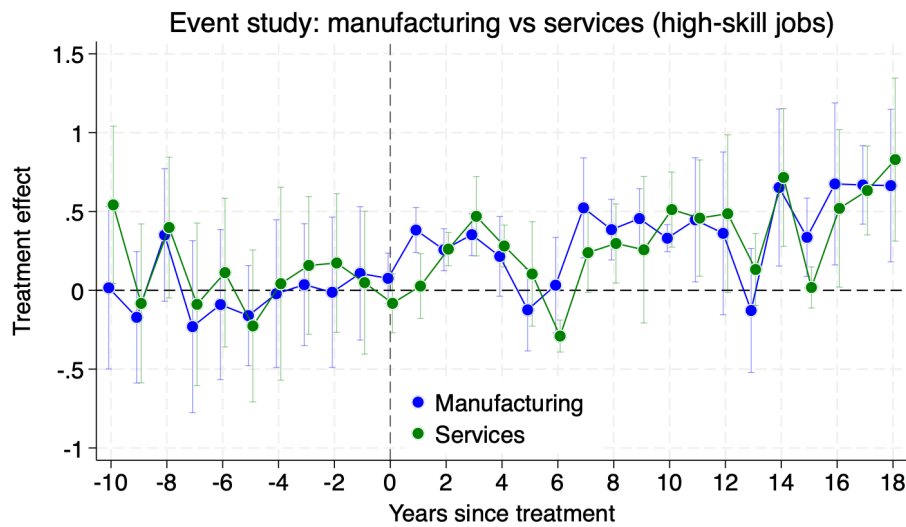


Figure 23: Long-run event-study effect of new factory mentions on demand for skilled labor. The dependent variable is the log of occupation keyword hits normalized by a baseline basket of lower-skill occupations. Estimates are obtained using the Borusyak, Jaravel, and Spiess imputation estimator on region-year newspaper data. Vertical bars report 95 percent confidence intervals clustered by region. Data source: author's calculations using Library of Congress *Chronicling America*.

The estimates show that new factory mentions are followed by an increase in advertisements for skilled labor. Importantly, the increase is not limited to skilled manufacturing occupations. Skilled service and white-collar occupations also rise, consistent with the idea that manufacturing generates demand for organizational and administrative tasks. The help-wanted evidence therefore supports the occupational mechanism emphasized in the paper: manufacturing expansion is associated not only with greater demand for production-related skills, but also with greater demand for the skilled service and white-collar occupations that help organize production.

D Model Details, Equilibrium, and Wedge Implementation

This appendix collects technical material that is not included in the main text. Section D.1 defines the competitive equilibrium. Section D.2 derives the household fertility and education rules used in the quantitative model. Section D.3 states the regularity conditions and proof structure for the terminal service AGP results. Section E.5 clarifies how the counterfactual economies with different manufacturing peaks are generated and explains the connection between changing the manufacturing demand weight w_m and introducing an explicit price or value added tax wedge.

D.1 Competitive equilibrium

A competitive equilibrium is a sequence of household choices,

$$\{C_{ij,t}, C_{i,t}, n_{i,t}, e_{i,t}\}_{i,j,t},$$

firm choices,

$$\{VA_{j,t}, Y_{j,t}, L_{s,j,t}, L_{u,j,t}, M_{a,j,t}, M_{m,j,t}, PS_{j,t}, HS_t, LS_t\}_{j,t},$$

producer-service task allocations,

$$\{x_{j,t}, z_{j,t}^*\}_{j,t},$$

prices,

$$\{p_{a,t}, p_{m,t}, p_{c,t}, w_{s,t}, w_{u,t}, p_{hs,t}, p_{ls,t}\}_t,$$

and aggregate states,

$$\{L_t, S_t, H_t, A_{j,t}\}_{j,t},$$

such that the following conditions hold.

First, households maximize utility subject to their budget and time constraints. Second, final-good producers choose value added, agricultural intermediates, manufacturing intermediates, and producer services to minimize costs. Third, producer-service tasks are assigned to the least-cost mode, implying the high-skill task shares characterized in Section 4. Fourth, all goods, labor, and intermediate-input markets clear. Fifth, aggregate states evolve according to the laws of motion for population, the skilled share, sectoral productivity, and codified organizational knowledge.

Manufacturing is the numeraire:

$$p_{m,t} = 1.$$

The skilled and unskilled labor-market clearing conditions are

$$S_t L_t = \sum_{j \in \{a, m, c\}} L_{s, j, t} + L_{s, hs, t},$$

and

$$(1 - S_t) L_t = \sum_{j \in \{a, m, c\}} L_{u, j, t} + L_{u, ls, t}.$$

Final-good market clearing requires gross output to equal final use plus intermediate use. For agriculture and manufacturing,

$$Y_{j, t} = C_{j, t} + \sum_{k \in \{a, m, c\}} M_{j, k, t}, \quad j \in \{a, m\}.$$

Consumer services are used as final demand, while producer services enter production as intermediate inputs through the composite $PS_{j, t}$. Producer-service markets clear when aggregate demand for high- and low-skill producer-service modes equals the corresponding supplies HS_t and LS_t .

D.2 Household fertility and education rules

This subsection derives the closed-form fertility and education rules used in the quantitative implementation. For a household of origin $i \in \{a, m, s\}$, the fertility-education part of the problem can be written as

$$\max_{n_{i, t}, e_{i, t}} (1 - \gamma) \log [1 - n_{i, t} (\tau_i + \tau_e(S_t) e_{i, t})] + \gamma \log n_{i, t} + \gamma \log h_{i, t},$$

where

$$h_{i, t} = \frac{e_{i, t} + \tau_h}{e_{i, t} + \tau_h + g_t}.$$

Here $i = a$ denotes agricultural origin, $i = m$ manufacturing origin, and $i = s$ service sector origin. This origin index is distinct from the final-good sector index $j \in \{a, m, c\}$, where c denotes consumer services.

Let

$$\chi_t \equiv \tau_e(S_t).$$

The first-order condition for fertility is

$$\frac{(1 - \gamma)(\tau_i + \chi_t e_{i, t})}{1 - n_{i, t}(\tau_i + \chi_t e_{i, t})} = \frac{\gamma}{n_{i, t}}.$$

Solving gives

$$n_{i, t} = \frac{\gamma}{\tau_i + \chi_t e_{i, t}}.$$

Substituting this expression into the household problem, the interior education choice solves

$$\frac{\partial \log h_{i,t}}{\partial e_{i,t}} = \frac{\chi_t}{\tau_i + \chi_t e_{i,t}}.$$

Using the human capital function,

$$\frac{\partial \log h_{i,t}}{\partial e_{i,t}} = \frac{g_t}{(e_{i,t} + \tau_h)(e_{i,t} + \tau_h + g_t)}.$$

Thus the interior education choice satisfies

$$\frac{g_t}{(e_{i,t} + \tau_h)(e_{i,t} + \tau_h + g_t)} = \frac{\chi_t}{\tau_i + \chi_t e_{i,t}}.$$

Solving for $e_{i,t}$ yields

$$e_{i,t}^* = \sqrt{\frac{g_t(\tau_i - \chi_t \tau_h)}{\chi_t}} - \tau_h.$$

Imposing non-negativity,

$$e_{i,t}^* = \max \left\{ \sqrt{\frac{g_t(\tau_i - \chi_t \tau_h)}{\chi_t}} - \tau_h, 0 \right\}.$$

The two closed-form rules used in the model are therefore

$$n_{i,t} = \frac{\gamma}{\tau_i + \chi_t e_{i,t}^*},$$

and

$$e_{i,t}^* = \max \left\{ \sqrt{\frac{g_t(\tau_i - \chi_t \tau_h)}{\chi_t}} - \tau_h, 0 \right\}.$$

These expressions make explicit the demographic channel. Higher growth g_t raises the return to education, which increases education investment per child. Higher education investment raises the time cost of children and lowers fertility.

D.3 Limiting service economy derivations

This appendix proves the propositions in Section 4.5. The main distinction from a standard three-sector structural-change model is that services enter the model in two ways. Consumer services are final goods purchased by households, while producer services are intermediate inputs used by all final-good sectors. Therefore, the long-run service economy should be interpreted as a statement about the composition of final demand, not as a statement that all employment is literally in consumer services. Even if final expenditure converges to consumer services, agriculture, manufacturing, and producer services may remain active through input-output linkages.

I proceed in four steps. First, I show that the non-homothetic CES demand system implies asymptotic concentration of final expenditure in consumer services under the standard long-run income and price restrictions emphasized by [Comin et al. \(2021\)](#). Second, I characterize the terminal input-output economy that supports this service final-demand limit. Third, I prove Proposition 1. Fourth, I prove Proposition 2 and then extend the logic to arbitrary finite transition paths with interior employment shares.

Long-run final demand. Let

$$E_{j,t} = p_{j,t} C_{j,t}$$

denote final expenditure on sector $j \in \{a, m, c\}$. The non-homothetic CES demand system in equation (8) implies the relative final-expenditure equation

$$\frac{E_{j,t}}{E_{k,t}} = \frac{\omega_j}{\omega_k} \left(\frac{p_{j,t}}{p_{k,t}} \right)^{1-\sigma} C_t^{(1-\sigma)(\varepsilon_j - \varepsilon_k)}. \quad (35)$$

This expression is the key demand-side object. It combines a long-run price effect, through relative prices, and a long-run income effect, through aggregate real consumption. This is the force emphasized by [Comin et al. \(2021\)](#).

The long-run service demand restriction is

$$C_t \rightarrow \infty, \quad (36)$$

$$\left(\frac{p_{j,t}}{p_{c,t}} \right)^{1-\sigma} C_t^{(1-\sigma)(\varepsilon_j - \varepsilon_c)} \rightarrow 0, \quad j \in \{a, m\}. \quad (37)$$

Condition (37) says that the income effect favoring consumer services dominates any long-run relative-price force favoring agriculture or manufacturing. In the empirically relevant case with $\sigma < 1$ and $\varepsilon_c > \varepsilon_m > \varepsilon_a$, this condition holds whenever relative prices do not grow fast enough to offset the income effect toward services.

Lemma 1: final demand is asymptotically concentrated in consumer services. Under (36)–(37),

$$\frac{E_{a,t}}{E_{c,t}} \rightarrow 0, \quad \frac{E_{m,t}}{E_{c,t}} \rightarrow 0. \quad (38)$$

Hence, if

$$b_{j,t} = \frac{E_{j,t}}{\sum_{k \in \{a, m, c\}} E_{k,t}}$$

denotes the final-expenditure share of sector j , then

$$(b_{a,t}, b_{m,t}, b_{c,t}) \rightarrow (0, 0, 1). \quad (39)$$

Proof. Set $k = c$ in (35). For $j \in \{a, m\}$,

$$\frac{E_{j,t}}{E_{c,t}} = \frac{\omega_j}{\omega_c} \left(\frac{p_{j,t}}{p_{c,t}} \right)^{1-\sigma} C_t^{(1-\sigma)(\varepsilon_j - \varepsilon_c)}.$$

The term multiplying ω_j/ω_c converges to zero by (37). Therefore $E_{j,t}/E_{c,t} \rightarrow 0$ for $j = a, m$. Since final-expenditure shares sum to one, this implies (39). $\square$

The terminal input-output economy. Lemma 1 does not imply that agriculture and manufacturing disappear from gross output or employment. They may remain active as intermediate-input sectors. To make this explicit, collect agriculture, manufacturing, consumer services, and producer services in the gross-output vector

$$y_t = (Y_{a,t}, Y_{m,t}, Y_{c,t}, PS_t)'$$

Let final demand be

$$f_t = (f_{a,t}, f_{m,t}, f_{c,t}, 0)'$$

Producer services have no final demand and enter only as intermediate inputs. Let A_t^{IO} denote the input-output matrix over these four production sectors. Gross output satisfies

$$y_t = A_t^{IO} y_t + f_t. \quad (40)$$

Suppose that

$$A_t^{IO} \rightarrow A^{IO,*}(H^*)$$

and that the limiting matrix is productive:

$$\rho(A^{IO,*}(H^*)) < 1, \quad (41)$$

where $\rho(\cdot)$ denotes the spectral radius. By Lemma 1, the limiting final-demand vector is proportional to consumer-service final demand. Therefore the terminal gross-output vector is

$$y^*(H^*) = (I - A^{IO,*}(H^*))^{-1} f_c^* e_c, \quad (42)$$

where $e_c = (0, 0, 1, 0)'$ and $f_c^* > 0$ is the scale of final consumer-service demand.

Equation (42) is the input-output counterpart of the service economy limit. Final demand is concentrated in consumer services, but the Leontief inverse determines the gross output of agriculture, manufacturing, and producer services required to support that final service demand. Thus two economies can share the same terminal final-demand vector while differing in the skill composition of producer services, the skilled share, the organizational frontier, and potentially the terminal gross-output and employment vectors if $A^{IO,*}$ depends on H^* .

Terminal skill market clearing. The terminal economy is indexed by the inherited organizational frontier H^* . Let

$$r = \frac{w_s}{w_u}$$

denote the skill premium. For a given H^* , the high-skill task share in producer services used by final sector j is

$$x_j(H^*, r) = 1 - \exp \left(- \frac{\phi(H^*) \max \left\{ 1 + \log \left(\frac{p_{ls}}{p_{hs}} \right), 0 \right\}}{\kappa_j} \right), \quad (43)$$

where

$$p_{hs} = \frac{w_s}{A_{hs}h_s}, \quad p_{ls} = \frac{w_u}{A_{ls}h_u}.$$

Holding r fixed, $x_j(H^*, r)$ is increasing in H^* . Holding H^* fixed, $x_j(H^*, r)$ is weakly decreasing in r , because a higher skill premium raises the relative cost of high-skill producer services.

Let $D_s(r; H^*)$ denote the terminal relative demand for skilled labor implied by value added production, producer-service task allocation, and the terminal input-output vector in (42). Let $S^{sup}(r)$ denote the terminal skilled labor supply generated by the skill choice equation:

$$S^{sup}(r) = \frac{1}{1 + \exp \left(- \frac{\beta_\pi \log r - c_0}{\sigma_{\text{skill}}} \right)}. \quad (44)$$

The terminal skill premium solves

$$D_s(r; H^*) = S^{sup}(r). \quad (45)$$

The following restrictions are sufficient for uniqueness and monotone comparative statics of the terminal skill equilibrium:

1. $D_s(r; H)$ is continuous and strictly decreasing in r .
2. $D_s(r; H)$ is weakly increasing in H , and strictly increasing in H whenever producer-service demand is positive.
3. $S^{sup}(r)$ is continuous and strictly increasing in r .
4. The endpoints cross:

$$\lim_{r \downarrow 0} D_s(r; H) > \lim_{r \downarrow 0} S^{sup}(r), \quad \lim_{r \uparrow \infty} D_s(r; H) < \lim_{r \uparrow \infty} S^{sup}(r).$$

These restrictions are primitive single-crossing conditions. CES technologies imply that relative demand for skilled labor falls with the skill premium. The logit skill supply equation implies that relative skill supply rises with the skill premium. A

higher H expands the set of producer-service tasks performed in high-skill mode and therefore shifts skilled-labor demand outward.

Lemma 2: the terminal skilled share is uniquely determined by H^* . Under the four restrictions above, for each $H^* > 0$, there exists a unique terminal skill premium $r^*(H^*)$ solving (45). The associated terminal skilled share

$$S^*(H^*) = S^{sup}(r^*(H^*))$$

is increasing in H^* .

Proof. Define the excess demand function

$$\Psi(r; H) = D_s(r; H) - S^{sup}(r).$$

By continuity of D_s and S^{sup} , Ψ is continuous in r . By the endpoint conditions, $\Psi(r; H)$ is positive for sufficiently low r and negative for sufficiently high r . Hence a solution exists by the intermediate value theorem. Since $D_s(r; H)$ is strictly decreasing in r and $S^{sup}(r)$ is strictly increasing in r , $\Psi(r; H)$ is strictly decreasing in r . Hence the solution is unique.

Now take $H' > H$. Since $D_s(r; H') \geq D_s(r; H)$ for all r , with strict inequality over the relevant range, the excess demand curve shifts upward. Because Ψ is strictly decreasing in r , the unique root satisfies

$$r^*(H') > r^*(H).$$

Since $S^{sup}(r)$ is strictly increasing in r ,

$$S^*(H') = S^{sup}(r^*(H')) > S^{sup}(r^*(H)) = S^*(H).$$

Thus $S^*(H)$ is increasing in H . $\square$

Proof of Proposition 1. By Lemma 1, final expenditure converges to consumer services. By (42), the limiting final-demand vector and the limiting input-output matrix determine the terminal gross-output vector. Hence the terminal production network is well defined even though agriculture, manufacturing, and producer services may remain active as intermediate-input sectors.

By Lemma 2, the inherited organizational frontier H^* selects a unique terminal skilled share

$$S^* = S(H^*), \quad S'(H) > 0.$$

It remains to characterize fertility, education, growth, and population on the terminal path.

In the terminal service final-demand economy, the limiting household origin is service origin. Therefore the fertility rule derived in Section D.2 gives

$$n^* = \frac{\gamma}{\tau_s + \tau_e(S^*)e^*}. \quad (46)$$

A terminal asymptotic growth path with finite population scale requires replacement fertility:

$$n_t \rightarrow 1.$$

Combining this condition with (46) gives

$$e^* = \frac{\gamma - \tau_s}{\tau_e(S^*)}. \quad (47)$$

The household education rule derived in Section D.2 is

$$e^* = \sqrt{\frac{g^*(\tau_s - \tau_e(S^*)\tau_h)}{\tau_e(S^*)}} - \tau_h.$$

Substituting (47) into this expression and solving for g^* gives

$$g^* = \frac{(\gamma - \tau_s + \tau_e(S^*)\tau_h)^2}{\tau_e(S^*)(\tau_s - \tau_e(S^*)\tau_h)}. \quad (48)$$

This is equation (32) in the main text.

Finally, on the terminal path, aggregate productivity growth is governed by the consumer-service branch of the productivity law:

$$g^* = H^*(S^*L^*)^{\varphi_c}.$$

Solving for L^* yields

$$L^* = \frac{1}{S^*} \left(\frac{g^*}{H^*} \right)^{1/\varphi_c}. \quad (49)$$

This is equation (33) in the main text.

Equations (47), (48), and (49), together with Lemma 2, establish that the terminal AGP is uniquely indexed by H^* . $\square$

A monotonicity corollary. Suppose that the education cost satisfies

$$\tau'_e(S) < 0$$

and that

$$\tau_s - \tau_e(S)\tau_h > 0$$

over the relevant range. Then $g^*(H)$ is increasing in H . Lemma 2 gives $S'(H) > 0$. Since $\tau_e'(S) < 0$, a higher H lowers the terminal education cost $\tau_e(S(H))$. Equation (48) is decreasing in $\tau_e(S^*)$ over the interior region. Hence

$$H_A^* > H_B^* \Rightarrow S_A^* > S_B^*, \quad e_A^* > e_B^*, \quad g_A^* > g_B^*.$$

This is the sense in which the inherited organizational frontier selects the skill intensity and growth rate of the mature service economy.

Proof of Proposition 2. Consider the three-period branching experiment in Figure 7. Both economies begin in agriculture with the same state (S_1, L_1, H_1) . Economy A passes through manufacturing in the intermediate period, while economy B moves directly into consumer services. In each intermediate period, one sector absorbs all labor.

By Proposition 1 and the monotonicity corollary,

$$g_A^* > g_B^* \iff H_A^* > H_B^*.$$

It remains to compare H_A^* and H_B^* .

In the limiting case $\tau_h \rightarrow 0$, the household education rule becomes

$$e_{i,t} = \sqrt{\frac{g_t \tau_i}{\tau_e(S_t)}}.$$

At the intermediate date, the two economies have the same initial S_1, L_1, H_1 , and education cost $\tau_e(S_1)$. If the intermediate sector is $q \in \{m, c\}$, the productivity-growth law gives

$$g_q = H_1(S_1 L_1)^{\varphi_q}.$$

Therefore intermediate education in sector q is

$$e_q = \sqrt{\frac{H_1(S_1 L_1)^{\varphi_q} \tau_q}{\tau_e(S_1)}}.$$

Because the intermediate period is fully specialized in one sector, current codification capacity is

$$\tilde{H}_q = \frac{e_q}{\kappa_q}.$$

Thus economy A , which passes through manufacturing, has intermediate codification capacity

$$\tilde{H}_A = \frac{1}{\kappa_m} \sqrt{\frac{H_1(S_1 L_1)^{\varphi_m} \tau_m}{\tau_e(S_1)}},$$

while economy B , which passes directly into consumer services, has intermediate codification capacity

$$\tilde{H}_B = \frac{1}{\kappa_c} \sqrt{\frac{H_1(S_1 L_1)^{\varphi_c} \tau_s}{\tau_e(S_1)}}.$$

The organizational frontier is irreversible:

$$H_{t+1} = H_t + \max\{\tilde{H}_t - H_t, 0\} = \max\{H_t, \tilde{H}_t\}.$$

Since both economies start from the same H_1 , the manufacturing route leaves a higher inherited frontier if and only if

$$\tilde{H}_A > \tilde{H}_B.$$

Using the two expressions above,

$$\frac{1}{\kappa_m} \sqrt{(S_1 L_1)^{\varphi_m} \tau_m} > \frac{1}{\kappa_c} \sqrt{(S_1 L_1)^{\varphi_c} \tau_s}.$$

Squaring both sides gives

$$(S_1 L_1)^{\varphi_m - \varphi_c} \frac{\tau_m}{\tau_s} > \left(\frac{\kappa_m}{\kappa_c} \right)^2.$$

Therefore

$$g_A^* > g_B^* \iff H_A^* > H_B^* \iff (S_1 L_1)^{\varphi_m - \varphi_c} \frac{\tau_m}{\tau_s} > \left(\frac{\kappa_m}{\kappa_c} \right)^2.$$

This proves Proposition 2. $\square$

Extension to arbitrary finite transition paths with interior employment shares. The three-period experiment gives a closed-form condition because each intermediate economy is fully specialized in one sector. The same logic extends to arbitrary finite transition paths with interior sectoral employment shares, but the extension must be stated in terms of inherited organizational frontiers rather than service only employment limits.

Let P denote a transition path before the economy reaches its terminal final-demand regime. Along path P , define current codification capacity as

$$\tilde{H}_t(P) = \bar{e}_t(P) \sum_{j \in \{a, m, c\}} s_{j,t}(P) \frac{1}{\kappa_j}. \quad (50)$$

The law of motion for codified organizational knowledge implies

$$H_{t+1}(P) = \max\{H_t(P), \tilde{H}_t(P)\}. \quad (51)$$

Therefore, for any finite transition path P ,

$$H^*(P) = \max \left\{ H_0, \sup_{t \in P} \tilde{H}_t(P) \right\}. \quad (52)$$

Thus the entire finite transition affects the terminal economy through the maximum effective codification capacity attained along the path.

Now consider two transition paths A and B with the same primitives and the same initial state. Suppose both paths satisfy the long-run final-demand condition

$$(b_{a,t}, b_{m,t}, b_{c,t}) \rightarrow (0, 0, 1),$$

so that final expenditure is asymptotically concentrated in consumer services. This does not imply that all employment is in consumer services. With input-output linkages, the terminal gross-output vector is

$$y^*(H^*) = \left(I - A^{IO,*}(H^*) \right)^{-1} f_c^* e_c.$$

Hence agriculture, manufacturing, and producer services may remain active as intermediate-input sectors even though final demand is concentrated in consumer services.

The correct general comparison is therefore the following. If two paths A and B share the same terminal final-demand regime and the same terminal input-output mapping except through the inherited frontier H^* , then the path with the higher inherited frontier is the path with the higher maximum codification capacity:

$$H^*(A) > H^*(B) \iff \sup_{t \in A} \left[\bar{e}_t(A) \sum_{j \in \{a,m,c\}} s_{j,t}(A) \frac{1}{\kappa_j} \right] > \sup_{t \in B} \left[\bar{e}_t(B) \sum_{j \in \{a,m,c\}} s_{j,t}(B) \frac{1}{\kappa_j} \right], \quad (53)$$

up to the common inherited frontier H_0 .

If, in addition, the terminal static equilibrium satisfies the monotone frontier property established above—namely, a higher H^* raises high-skill producer-service task use, shifts out skilled-labor demand, raises the terminal skilled share, and lowers the effective cost of education—then

$$H^*(A) > H^*(B) \implies S_A^* > S_B^*, \quad e_A^* > e_B^*, \quad g_A^* > g_B^*.$$

This is the general-path version of the manufacturing mechanism. Manufacturing raises long-run service economy growth if and only if its presence along the transition raises the organizational frontier inherited by the terminal economy. The three-period result in Proposition 2 is the closed-form special case in which the comparison

of inherited frontiers can be reduced to

$$(S_1 L_1)^{\varphi_m - \varphi_c} \frac{\tau_m}{\tau_s} > \left(\frac{\kappa_m}{\kappa_c} \right)^2.$$

This extension also clarifies the role of input-output linkages. Long-run final demand may be identical across paths, but terminal employment need not be. If $A^{IO,*}(H^*)$ depends on the inherited organizational frontier, then paths with different H^* can have different terminal gross-output and employment vectors. If $A^{IO,*}$ is common across paths, then the terminal gross-output vector is common, but the skill composition of producer services, the skilled share, education, fertility, and terminal growth can still differ through H^* .

D.4 Historical examples of producer-service codification

This appendix gives historical examples that clarify the interpretation of the codification friction κ_j and the organizational state H_t . The examples are not used to identify or calibrate the model. They illustrate the economic content of the mechanism: producer-service tasks can be performed informally in low-skill mode, or they can be reorganized into codified routines that require skilled workers and can be reused across firms and sectors.

The key distinction is between a function and the mode in which that function is performed. Work organization, quality control, inventory management, accounting, procurement, and personnel management exist in many economies. In low-skill mode, these functions are carried out through local experience, tacit knowledge, and informal supervision. In high-skill mode, the same functions are transformed into records, procedures, metrics, job categories, and decision rules. Manufacturing and industrial procurement are natural settings for this transformation because many of the relevant tasks are repeated, measurable, and costly to perform informally. Once codified, however, the resulting routines can travel into service organizations such as department stores, hospitals, banks, retailers, offices, and universities.

Table 16 summarizes the mapping from the model to several historical examples.

Table 16: Examples of producer-service tasks moving from low-skill to high-skill mode

Producer-service task	Low-skill mode	Industrial codification	Later service sector use
Work organization	Foreman judgment, local rules of thumb, informal supervision	Motion study, workflow design, timing of tasks, written procedures, training systems	Department stores, offices, hospitals, and other service organizations use workflow design, scheduling, clerical organization, and personnel practices
Quality control	Inspection by eye, informal rejection of defective output	Testing protocols, defect records, statistical process control, control charts	Clinical laboratories and hospitals use control charts and statistical quality-control routines to monitor measurement processes
Accounting and managerial control	Bookkeeping and transaction recording	Cost accounting, budgeting, return-on-investment calculations, divisional reporting, performance metrics	Retail chains, department stores, banks, hospitals, universities, and other multi-unit service organizations use budgeting and management-control systems
Inventory, procurement, and logistics	Visual stock checks, ad hoc ordering, relationship-based purchasing	Stock-control systems, procurement planning, supplier coordination, delivery scheduling	Retail chains, wholesalers, hospitals, restaurants, and distribution firms use inventory and logistics routines
Personnel and training	Informal apprenticeship, personal hiring, on-the-job learning	Job analysis, hiring rules, training manuals, personnel departments, performance evaluation	Department stores, hotels, hospitals, offices, and other service organizations use formal hiring and training systems

Work organization: from factory motion study to department stores. A first example is work organization. In low-skill mode, the task of organizing work is performed by foremen and supervisors using experience and local knowledge. Manufacturing made this task easier to codify because factory operations were repeated and measurable: worker motions, delays, output, and defects could be observed, timed, compared, and redesigned. The scientific-management tradition transformed work organization into a high-skill producer-service function involving motion study, workflow design, written procedures, and training systems.

Lillian Gilbreth provides a useful historical example of diffusion across sectors. Frank and Lillian Gilbreth developed motion-study methods within the scientific-management tradition, initially oriented toward industrial work. Historical accounts emphasize that Lillian Gilbreth later adapted motion study and industrial psychology to service and household settings, including department-store employees and consumers (Graham, 1992, 1999). In particular, her work at Macy's between 1925 and 1928 applied scientific-management and personnel-management ideas to hiring, training, employee handling, clerical organization, and the organization of work within the department store (Graham, 2000). This is the mechanism in the model: manufacturing and industrial work make a producer-service task easier to codify, and the resulting routine can later be used by service organizations.

Quality control: from industrial process control to clinical laboratories. A second example is quality control. In low-skill mode, quality is monitored through direct inspection or the judgment of experienced workers. Manufacturing made quality problems measurable because defects, tolerances, downtime, and output variation could be recorded. This gave rise to statistical process control. Walter Shewhart's work at Bell Laboratories in the 1920s introduced the control chart, and his later book formalized statistical control as a method for monitoring variation in production processes (Shewhart, 1931; Best and Neuhauser, 2006).

Once codified, the same routine was portable outside manufacturing. Clinical laboratories later adopted statistical quality-control methods to monitor the reliability of laboratory measurements. Levey and Jennings introduced statistical quality-control charts for clinical laboratories in 1950, adapting the logic of process control to repeated measurement in medical testing (Levey and Jennings, 1950; Westgard et al., 1981). The task changed from informal inspection to a high-skill routine based on records, thresholds, and statistical decision rules. This example maps directly into the model: codified organizational knowledge H_t makes it easier for a service organization to adopt a high-skill version of a task that was first formalized in an industrial setting.

Accounting and managerial control: from industrial cost control to retail management. A third example is accounting and managerial control. In low-skill mode,

accounting records transactions and keeps the books. Large industrial firms and industrial procurement systems created a different problem: managers needed to compare plants, products, inventories, costs, divisions, and capital allocations. This transformed accounting into a high-skill producer-service activity: cost accounting, budgeting, return-on-investment analysis, divisional reporting, and management control. The historical literature on management accounting emphasizes the development of cost accounting and managerial control in large industrial and transportation enterprises, the influence of scientific management, and the later use of formal budgeting and performance measures in large corporations such as DuPont and General Motors (Johnson, 1975; Kaplan, 1984; Chandler, 1977).

The same managerial routines later became portable tools for service firms. James O. McKinsey helped professionalize budgeting and managerial accounting as a consulting service through books such as McKinsey (1922) and McKinsey (1924). In 1935, McKinsey completed a management consulting engagement for Marshall Field’s, a large Chicago department-store chain, and then became its chairman and chief executive (Flesher, 1985). The relevant diffusion is not a machine or a product, but a producer-service routine: a way of measuring, budgeting, and managing a complex organization. In the language of the model, management accounting is a task that moves from bookkeeping in low-skill mode to codified high-skill managerial control.

E Calibration Details and Counterfactual Implementation

This appendix provides additional details on the numerical calibration and clarifies the interpretation of the counterfactual variation in manufacturing peaks.

E.1 Numerical procedure

The model is simulated at an annual frequency and calibrated to the U.S. transition over 1850–2019. The simulation horizon is longer than the data window. The model date corresponding to 1850 is fixed at $t_0 = 60$, and all simulated moments are evaluated after applying this fixed calendar alignment. Gross-output input shares are interpolated so that the modern input-output structure is reached in calendar year 2025.

Let m^{data} denote the vector of empirical moments and $m^{model}(\Theta)$ the corresponding model moments. The calibrated parameter vector minimizes the loss

$$\mathcal{L}(\Theta) = \frac{1}{N} \sum_{\ell=1}^N \left(\frac{m_{\ell}^{model}(\Theta) - m_{\ell}^{data}}{\max\{|m_{\ell}^{data}|, \underline{s}_{\ell}\}} \right)^2,$$

for share and rate moments. Strictly positive level moments, such as normalized GDP per capita and the agriculture-to-service fertility ratio, are matched in logs. The constants $\underline{s}_\ell$ are moment-specific scale floors used to avoid overweighting moments close to zero.

The numerical minimization uses a global dual-annealing search followed by a Powell local search. I add small regularization penalties for high-frequency oscillations in the skill premium and in the aggregate skilled share. These penalties are not targeted economic moments; they rule out parameter vectors that fit isolated moments through implausibly volatile transition paths.

The half-saturation parameter in the codification frontier, $H_{1/2}$, is normalized relative to the simulated scale of codified organizational knowledge accumulation. For each candidate parameter vector, I first run a probe simulation with a fixed input value of $H_{1/2}$. I then set $H_{1/2}$ equal to the average of the final observations of the probe H_t path, subject to lower and upper bounds. This prevents the arbitrary units of H_t from mechanically determining the location of the frontier.

E.2 Calibrated parameters and targeted moments

The internally calibrated parameter vector is

$$\Theta = \left(\gamma, \tau_a, \varphi_a, \varphi_m, \varphi_c, g^{GO}, \kappa_a, \kappa_m, \kappa_c, \eta_H, c_{col}, \alpha_a, \alpha_m, \alpha_c, \varepsilon_a, \varepsilon_c, w_a, w_c \right).$$

The manufacturing demand weight is fixed at $w_m = 3.5$. Manufacturing is the numeraire and $\varepsilon_m = 1$.

The targeted moments are: sectoral employment shares in agriculture, manufacturing, and services; the peak manufacturing employment share; the fertility path; the agriculture-to-service fertility ratio; normalized GDP per capita; long-run value added productivity growth in agriculture, manufacturing, and consumer services; the aggregate skilled employment share; within-value added skilled employment shares by sector; high-skill employment shares in producer-service inputs by using sector; and the aggregate change in the high-skill producer-service share between 1960 and 2019. The calibration is overidentified: the number of targeted moments exceeds the number of internally calibrated parameters.

E.3 High-skill producer-service inputs by final sector

This subsection describes the construction of the high-skill producer-service moments used in the calibration. These moments discipline the sector-specific codification parameters κ_j , which determine how easily producer-service tasks used by final sector $j \in \{a, m, c\}$ can be reorganized into high-skill form.

The object I need is not the average education of all workers in agriculture, manufacturing, or consumer services. The model distinguishes final-good production from

the producer-service tasks used as intermediate inputs. The relevant empirical counterpart is therefore the education intensity of producer-service occupations employed within each broad final sector. In the model, these occupations correspond to managerial, professional, technical, administrative, accounting, engineering, planning, and coordination tasks.

I construct the U.S. moments using the 2019 IPUMS USA Census sample. I classify workers into broad industries using the 1950 Census industry classification. Agriculture includes agriculture, forestry, and fisheries. Manufacturing includes durable and nondurable manufacturing. Consumer services include retail, personal services, entertainment, education, health, welfare, religious services, and other consumer-facing service industries. I then identify producer-service occupations using the 1950 Census occupation classification. The baseline definition includes occupations that clearly perform professional, technical, managerial, or organizational tasks, such as accountants, architects, engineers, lawyers, statisticians, personnel workers, technicians, purchasing agents, inspectors, insurance agents, real-estate agents, and related business-facing occupations. The replication files report the exact industry and occupation codes.

For each broad final sector j , I compute the weighted share of producer-service workers with at least one year of college:

$$HSPS_j = \frac{\sum_{n \in j} \omega_n \mathbb{1}\{\text{PS occupation}_n = 1\} \mathbb{1}\{\text{college}_n = 1\}}{\sum_{n \in j} \omega_n \mathbb{1}\{\text{PS occupation}_n = 1\}},$$

where ω_n is the person weight. These are the empirical counterparts of the high-skill shares within producer-service inputs used by agriculture, manufacturing, and consumer services in the model. The resulting U.S. moments are

$$HSPS_a = 0.393, \quad HSPS_m = 0.653, \quad HSPS_c = 0.529.$$

Thus, producer-service occupations employed in manufacturing are substantially more college-intensive than producer-service occupations employed in agriculture or consumer services. This ranking is the key moment used to discipline the ordering of codification resistance in the model. The calibrated model chooses $\kappa_m < \kappa_c < \kappa_a$, so manufacturing related producer-service tasks are easiest to reorganize into high-skill form.

I also examine whether this pattern is specific to the United States. Using IPUMS International, I construct the same object for 298 census samples spanning countries at very different stages of development, including low, middle and high income countries. I then repeat the exercise by classifying industries into agriculture, manufacturing, and consumer services using the harmonized IPUMS industry variable. I classify producer-service occupations using ISCO major groups corresponding to managerial, professional, and technical occupations.¹⁰ I then compute college shares separately by broad sector and occupation type. Figure 24 reports the resulting averages for

10. This corresponds to ISCO code 3 or less.

agriculture, manufacturing, and consumer services. Producer-service occupations are much more college-intensive than other occupations in every broad sector. Among producer-service occupations, the college share is also higher in manufacturing than in consumer services or agriculture. This supports the interpretation of the U.S. calibration moments: the relevant margin is not that all manufacturing workers are uniformly more educated, but that manufacturing uses a set of producer-service occupations that is especially skill intensive.

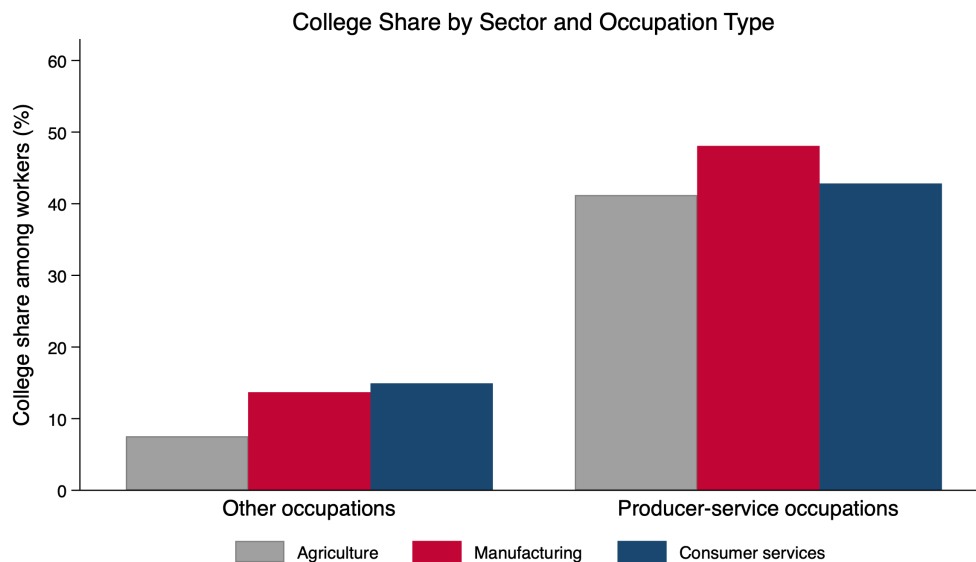


Figure 24: College share by sector and occupation type in IPUMS International. The figure uses 298 census samples covering economies across the development spectrum. Producer-service occupations are defined using managerial, professional, and technical occupation groups. Bars report average college shares separately for other occupations and producer-service occupations within agriculture, manufacturing, and consumer services.

E.4 Untargeted producer-service response

As an external validation of the producer-service block, I compare the model to a final-demand-shock exercise in the 2015 OECD ICIO table. In the data, I compute the service output induced by a \$100 final-demand shock to agriculture, manufacturing, and consumer services. In the model, I apply the analogous shock to the calibrated economy and compute the high-skill share of induced producer-service output. This moment is not directly targeted in the calibration.

E.5 Interpreting the manufacturing demand counterfactual

The counterfactuals in the main text vary the manufacturing demand shifter w_m to generate economies with different peak manufacturing employment shares. This is a reduced-form way to change the relative pull of manufacturing during the transition

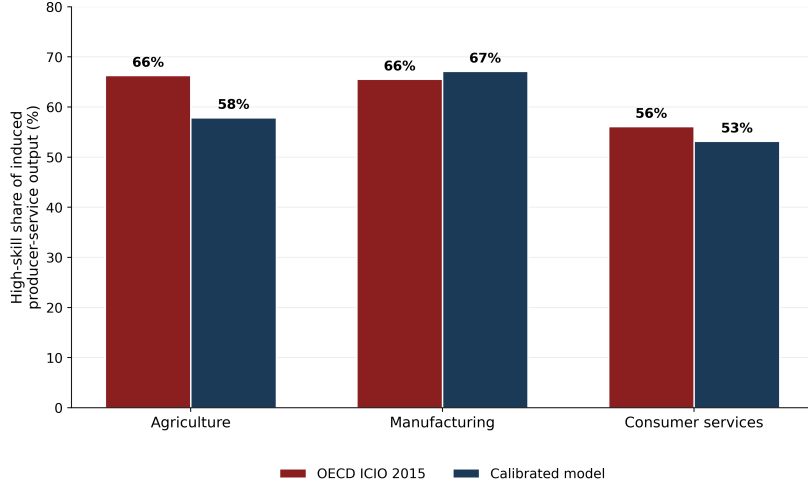


Figure 25: High-skill share of producer-service output induced by a \$100 final-demand shock. The data exercise uses Leontief total requirements from the 2015 OECD ICIO table; the model exercise applies the same shock to the calibrated economy.

while holding technology, fertility and education preferences, and production primitives fixed.

This counterfactual has an equivalent price-wedge representation. Final demand over agriculture, manufacturing, and consumer services implies relative demand terms of the form

$$K_a = \left(\frac{p_a}{p_m} \right)^{1-\sigma} \Upsilon^{(1-\sigma)(k_a-1)} \frac{w_a}{w_m^{k_a}}, \quad K_c = \left(\frac{p_c}{p_m} \right)^{1-\sigma} \Upsilon^{(1-\sigma)(k_c-1)} \frac{w_c}{w_m^{k_c}},$$

where $k_j \equiv \varepsilon_j / \varepsilon_m$. Changing the manufacturing demand weight from w_m to $f_m w_m$ therefore changes the relative demand terms to

$$K_a(f_m) = K_a(1) f_m^{-k_a}, \quad K_c(f_m) = K_c(1) f_m^{-k_c}.$$

Now consider an economy with the original w_m , but in which households face effective final-good prices

$$\tilde{p}_{a,t} = (1 + \tau_a) p_{a,t}, \quad \tilde{p}_{m,t} = p_{m,t}, \quad \tilde{p}_{c,t} = (1 + \tau_c) p_{c,t}.$$

The wedge economy generates the same relative demand schedules as the $f_m w_m$ economy if

$$(1 + \tau_a)^{1-\sigma} = f_m^{-k_a}, \quad (1 + \tau_c)^{1-\sigma} = f_m^{-k_c}.$$

Equivalently,

$$\boxed{1 + \tau_a = f_m^{-k_a/(1-\sigma)}, \quad 1 + \tau_c = f_m^{-k_c/(1-\sigma)}}.$$

Thus the w_m counterfactual can be interpreted either as a manufacturing demand shifter or as an equivalent system of final-good wedges. The two representations are

algebraically identical for household final-demand allocations. The purpose of varying w_m is therefore not to assign structural content to tastes, but to provide a parsimonious parameterization of a relative-price distortion that changes the size of the manufacturing phase.

E.6 Construction of input-output coefficients

This subsection describes how I construct the modern input-output coefficients used in the calibration. The model production function for each final-good sector $j \in \{a, m, c\}$ uses value added, agricultural intermediates, manufacturing intermediates, and producer-service intermediates:

$$Y_{j,t} = A_{go,j,t} VA_{j,t}^{\theta_{j,t}} M_{a,j,t}^{\gamma_{a,j,t}} M_{m,j,t}^{\gamma_{m,j,t}} PS_{j,t}^{\gamma_{ps,j,t}}, \quad \theta_{j,t} + \gamma_{a,j,t} + \gamma_{m,j,t} + \gamma_{ps,j,t} = 1.$$

The modern - 2025 - coefficients are constructed from the BEA Use Table, after redefinitions, at purchasers' prices. I use the sector-level table and map BEA industries into the model sectors as follows. Agriculture is BEA sector 11, manufacturing is BEA sector 31G, and consumer services are the sum of educational services, health care, and social assistance; arts, entertainment, recreation, accommodation, and food services; and other services except government, corresponding to BEA sectors 6, 7, and 81. Producer-service inputs are the sum of transportation and warehousing, information, finance, insurance, real estate, rental, and leasing, and professional and business services, corresponding to BEA sectors 48TW, 51, *FIRE*, and *PROF*.

For each destination sector j , I compute gross output Y_j and four raw shares:

$$\theta_j^{raw} = \frac{VA_j}{Y_j}, \quad \gamma_{a,j}^{raw} = \frac{U_{A,j}}{Y_j}, \quad \gamma_{m,j}^{raw} = \frac{U_{M,j}}{Y_j}, \quad \gamma_{ps,j}^{raw} = \frac{U_{PS,j}}{Y_j},$$

where $U_{A,j}$ is the use of agricultural inputs by destination sector j , $U_{M,j}$ is the use of manufacturing inputs, and $U_{PS,j}$ is the use of producer-service inputs. Since the model abstracts from residual input categories such as mining, utilities, construction, government, scrap, and noncomparable imports, I normalize the four model-relevant shares to sum to one:

$$(\theta_j, \gamma_{a,j}, \gamma_{m,j}, \gamma_{ps,j}) = \frac{(\theta_j^{raw}, \gamma_{a,j}^{raw}, \gamma_{m,j}^{raw}, \gamma_{ps,j}^{raw})}{\theta_j^{raw} + \gamma_{a,j}^{raw} + \gamma_{m,j}^{raw} + \gamma_{ps,j}^{raw}}.$$

For agriculture, the BEA table gives

$$\theta_a^{raw} = 0.394, \quad \gamma_{a,a}^{raw} = 0.295, \quad \gamma_{m,a}^{raw} = 0.222, \quad \gamma_{ps,a}^{raw} = 0.051.$$

After normalizing the four model-relevant components, the coefficients are

$$\theta_a = 0.410, \quad \gamma_{a,a} = 0.307, \quad \gamma_{m,a} = 0.230, \quad \gamma_{ps,a} = 0.053.$$

The lesson is that openness is most beneficial when it supports, rather than displaces, the manufacturing phase. Comparative advantage in manufacturing makes trade less likely to generate premature movement into low-skill services.

F.5 Development gaps and timing

Finally, I vary both the timing of trade opening and the development gap between Home and the rest of the world. Figure 32 reports two heatmaps. The left panel reports total discounted gains from trade. The right panel reports the long-run dynamic gain, measured by the final value of $W_t^{open} - W_t^{protected}$.

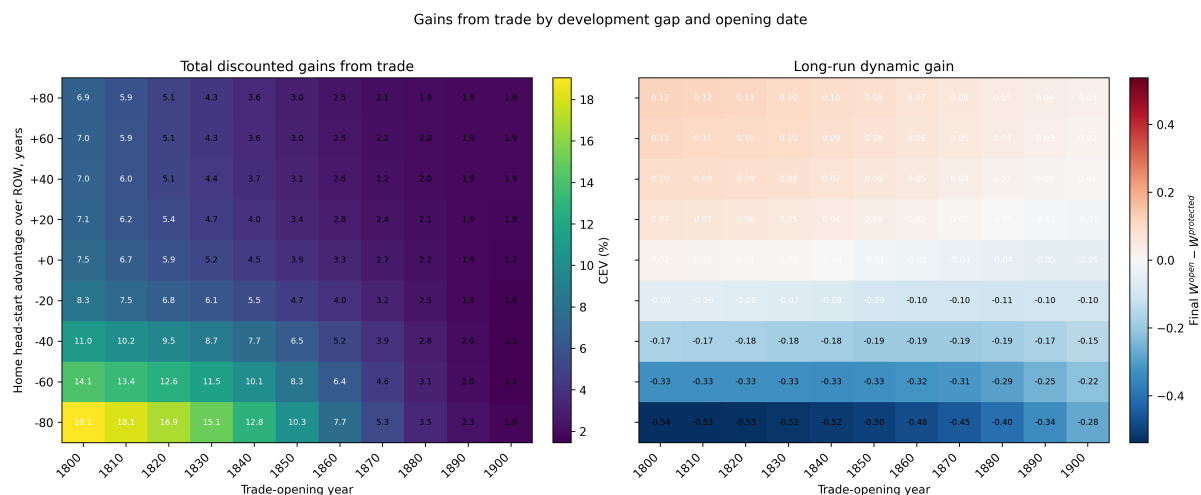


Figure 32: Gains from trade by development gap and opening date. Positive values on the vertical axis mean that Home is more advanced than the rest of the world. Negative values mean that Home opens to a more advanced partner.

The two panels reveal a sharp distinction between total and dynamic gains. Total discounted gains are largest when Home opens early to a more advanced partner. This reflects the large static benefits from cheaper traded goods. Long-run dynamic gains display the opposite pattern. They are positive when Home is more advanced than its partner, and negative when Home opens to a more advanced partner.

This asymmetry follows directly from the model mechanism. A less advanced economy opening to a more advanced partner imports cheaper manufactures, contracts domestic manufacturing, and enters services earlier. This raises consumption in the short run but weakens the domestic accumulation of high-skill producer-service capabilities. By contrast, an advanced economy opening to a less advanced partner preserves its manufacturing and producer-service base while still benefiting from trade. In that case, the long-run dynamic welfare effect can remain positive.

These results should however not be read as an argument for autarky. This exercise abstracts from technology diffusion, foreign direct investment, imported capital goods, and learning through exporting. Instead, we isolate one force: when manufacturing builds the human capital and organizational foundations of high-growth services,

trade policy affects development not only through prices, but also through the type of service economy that emerges.